



Enterprise HRMS — UI Navigation & Feature Testing Guide

Note

How Navigation Works in This Guide

- **The Left Sidebar:** This is the primary vertical menu on the left side of your screen. It lists main feature areas (such as **People**, **Attendance**, **Leave Management**, and **Holiday**).
- **Nested Menus & Tabs:** Some instructions tell you to click a main menu item first, then select a specific sub-item or tab across the top of the page (for example: *Click **Leave Management** in the left sidebar, then click the **Requests** tab*).
- **Technical Page Addresses:** You will see technical addresses in parentheses throughout this guide, such as `(/employees/directory)` or `(/leave)`. These are provided strictly as reference paths for QA testers and software developers cross-checking against the code. You do **not** need to type these addresses — simply follow the click-by-click instructions.



Local Environment & Access

- **Dev Server URL:** `http://localhost:5174/` (or `http://localhost:5173/`)
- **Framework:** React 19 + Vite + MUI v6 + Zod Validation



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1. 🛡️ Authentication & Account Setup

1.1 Sign Up / User Registration

- **Testing Steps:**
 1. Open your web browser and go to the HRMS sign-up page.
 2. Enter **Company Name** (e.g. `Creator Tech`) — verify **Workspace URL** auto-generates as `creatertech` (clean alphanumeric, without hyphens or spaces).
 3. Select **Team Size** from the dropdown menu (e.g. `11 - 50 employees`).
 4. Enter **First Name** and **Last Name**.
 5. Enter **Work Email** (must be a valid business domain email; personal domains like `gmail.com` are rejected on backend submit).
 6. Enter **Password** and **Confirm Password** (must match, 8-12 chars, containing uppercase, lowercase, digit, and special character).
 7. Select **Country Code** and enter **Phone Number** (digits only, validated per country format).
 8. Click **Create Account**.

✅ Success Cases

ACTION	STEPS	EXPECTED RESULT
Workspace URL Auto-Generation	Type <code>Creator Tech</code> into Company Name field	<code>workspaceSlug</code> field automatically populates with <code>creatertech</code> . Real-time availability indicator displays <code>✓ "creatertech" is available</code> in green (<code>#10B981</code>).
Pick Suggested Slug	Type a taken slug name, click any suggestion chip (e.g. <code>creatertech1</code>)	Workspace URL field updates to <code>creatertech1</code> , availability indicator updates to <code>✓ "creatertech1" is available</code> .
Successful Account Registration	Fill all required fields with valid data & click Create Account	Submit button shows loading spinner (<code>loading={true}</code>). Backend returns <code>201 Created</code> with message <code>"Company registered successfully"</code> . Redux state <code>isRegisterSuccess</code> triggers navigation to <code>/check-email</code> with state <code>{ email: "user@domain.com" }</code> .
Already Authenticated Redirect	Navigate directly to <code>/signup</code> while logged in	<code>GuestGuard</code> intercepts request (<code>hasSession === true</code>) and redirects instantly to <code>/dashboard</code> .

⚠️ Validation Errors to Test

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE
Company Name	Leave empty or enter 1 character	<code>"Company name must be at least 2 characters"</code>	Frontend Zod (<code>signup.schema.ts</code>)
Company Name	Enter text > 200 characters	<code>"Company name cannot exceed 200 characters"</code>	Frontend Zod (<code>signup.schema.ts</code>)
Workspace URL	Clear field or enter < 3 characters	<code>"Workspace URL must be at least 3 characters"</code>	Frontend Zod (<code>signup.schema.ts</code>)

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE
Workspace URL	Enter text > 63 characters	"Workspace URL cannot exceed 63 characters"	Frontend Zod (<code>signup.schema.ts</code>)
Workspace URL	Enter non-alphanumeric/hyphen characters	"Only lowercase letters, numbers, and hyphens. Cannot start or end with a hyphen."	Frontend Zod (<code>signup.schema.ts</code>)
Workspace URL	Enter reserved name (e.g. <code>admin</code> , <code>api</code> , <code>app</code>)	"This workspace name is reserved. Please choose another."	Backend DTO (<code>common.validator.ts</code>)
Team Size	Unselected dropdown	"Please select your team size"	Frontend Zod (<code>signup.schema.ts</code>)
First Name	Leave empty or enter 1 character	"First name must be at least 2 characters"	Frontend Zod (<code>signup.schema.ts</code>)
First Name	Enter text > 100 characters	"First name cannot exceed 100 characters"	Frontend Zod (<code>signup.schema.ts</code>)
Last Name	Leave empty or enter 1 character	"Last name must be at least 2 characters"	Frontend Zod (<code>signup.schema.ts</code>)
Last Name	Enter text > 100 characters	"Last name cannot exceed 100 characters"	Frontend Zod (<code>signup.schema.ts</code>)
Work Email	Leave blank	"Email is required"	Frontend Zod (<code>signup.schema.ts</code>)
Work Email	Invalid email format (e.g. <code>user@com</code>)	"Please enter a valid email"	Frontend Zod (<code>signup.schema.ts</code>)
Work Email	Personal email domain (e.g. <code>gmail.com</code> , <code>yahoo.com</code>)	"Please use your work email address. Personal email providers are not allowed."	Backend DTO (<code>common.validator.ts</code>)
Password	Leave blank	"Password is required"	Frontend Zod (<code>signup.schema.ts</code>)
Password	Less than 8 characters	"Password must be at least 8 characters"	Frontend Zod (<code>signup.schema.ts</code>)
Password	Missing uppercase, lowercase, digit, or special char	"Password must contain at least one uppercase letter" / "Password must contain at least one number" / "Password must contain at least one special character"	Backend DTO (<code>common.validator.ts</code>)
Password	Exceeds 12 characters	"Password must not exceed 12 characters"	Backend DTO (<code>auth.dto.ts</code>)
Confirm Password	Leave blank	"Confirm password is required"	Frontend Zod (<code>signup.schema.ts</code>)
Confirm Password	Does not match Password	"Passwords do not match"	Frontend Zod (<code>signup.schema.ts</code>)
Phone Number	Contains letters or special characters	"Phone number must contain digits only"	Frontend Zod (<code>signup.schema.ts</code>)

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE
Phone Number	Exceeds 10 digits	"Phone number cannot exceed 10 digits"	Frontend Zod (<code>signup.schema.ts</code>)
Phone Number	Number invalid for selected country code	"Invalid phone number for the selected country"	Frontend Zod (<code>signup.schema.ts</code> / <code>libphonenumber-js</code>)

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR
Workspace URL Already Taken (409)	Slug already registered in database (<code>workspaceSlugExists</code>)	Real-time status shows <code>Already taken. Try:</code> followed by up to 3 suggestion chips. Red warning text displayed: <code>"Please choose an available workspace URL before continuing."</code> . Create Account button remains disabled (<code>disabled={slugAvailable === false}</code>).
Duplicate Email Registration (409)	Register with an email already in database	Backend rejects with <code>409 Conflict</code> . Top-right toast notification pops up displaying: <code>"Email already registered"</code> .
Slug Conflict on Submit (409)	Submit registration when workspace slug is claimed concurrently	Backend rejects with <code>409 Conflict</code> . Top-right toast notification pops up displaying: <code>"Workspace URL \"{workspaceSlug}\" is already taken."</code> .
Backend DTO Validation Failure (400)	Field fails backend DTO (e.g. personal email or password > 12 chars)	Backend rejects with <code>400 Bad Request</code> . Redux Saga helper <code>extractErrorMessage()</code> extracts <code>errors[0].message</code> and top-right toast notification displays the specific field error (e.g. <code>"Please use your work email address. Personal email providers are not allowed."</code>).
Slug Check API Failure	Backend server offline or network error during <code>checkSlug()</code> GET call	Real-time status displays orange warning: <code>Could not check availability. Please try again.</code>
Server Connection Failure (500/Network)	Backend offline or unhandled database exception	Top-right toast notification displays Axios error message (<code>"Registration failed"</code> or <code>"Something went wrong"</code>).

🔑 Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS?	RESTRICTIONS / WHAT THEY SEE INSTEAD	CODE REFERENCE
Unauthenticated / Guest	✅ Yes	Full access to <code>/signup</code> page and workspace creation form.	<code>AppRoutes</code> / <code>GuestGuard</code>
<code>ORG_ADMIN</code>	❌ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>
<code>HR_ADMIN</code>	❌ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>
<code>BRANCH_ADMIN</code>	❌ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>

ROLE SLUG	CAN ACCESS?	RESTRICTIONS / WHAT THEY SEE INSTEAD	CODE REFERENCE
LEADERSHIP	✗ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>
MANAGER	✗ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>
PRODUCT_MANAGER	✗ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>
EMPLOYEE	✗ No	Intercepted by <code>GuestGuard</code> (<code>hasSession === true</code>). Instantly redirected to <code>/dashboard</code> .	<code>GuestGuard.tsx:L25-L27</code>

1.2 Login Page

- **Route:** Login screen (`/login`)
- **Testing Steps:**
 1. Open your web browser and go to the HRMS login page.
 2. Enter your registered **Company Email** (e.g., `admin@company.com`). Notice the 500ms debounced email check: if workspace branding exists, the company logo and name appear dynamically above the heading.
 3. Enter your **Password**.
 4. (Optional) Check **Remember this device** to issue a 30-day persistent session token.
 5. Click **Sign In**.
 6. Observe post-login redirect priority:
 - **Priority 1:** If `requiresPasswordReset` is true $\rightarrow$ redirected to `/auth/change-password` .
 - **Priority 2:** If `onboardingCompleted` is false $\rightarrow$ redirected to `/onboarding` .
 - **Priority 3:** Role-based redirect (`ORG_ADMIN` $\rightarrow$ `/dashboard` , `HR` $\rightarrow$ `/hr/dashboard` , `EMPLOYEE` $\rightarrow$ `/employee/dashboard`).

✓ Success Cases

ACTION	STEPS	EXPECTED RESULT
Workspace Branding Auto-Discovery	Type a valid registered email into Company Email field & pause 500ms	Input displays circular loading indicator (<code>CircularProgress</code>). Upon response, heading updates to Company Name, workspace logo displays, and subtext updates to <code>"Sign in to your workspace"</code> .
SSO Email Detection	Type an email belonging to an SSO-enabled domain	Password field hides automatically. Alert displays: <code>"Continue with [provider]"</code> or <code>"SSO is enabled for this account."</code> .
Successful Authentication (Admin)	Enter valid credentials for <code>ORG_ADMIN</code> & click Sign In	<code>accessToken</code> & <code>refreshToken</code> saved in <code>localStorage</code> . Redirects to <code>/dashboard</code> .
First Login Workspace Setup Redirect	Log in with a newly registered account where <code>onboardingCompleted === false</code>	Priority redirect triggers: User is redirected to <code>/onboarding</code>

ACTION	STEPS	EXPECTED RESULT
Mandatory Password Change Redirect	Log in with an admin-invited account where <code>requiresPasswordReset === true</code>	Priority redirect triggers: User is redirected to <code>/auth/change-password</code> with <code>replace: true</code> .
Remember Device Token	Check Remember this device checkbox before submitting	<code>rememberDevice: true</code> is sent in payload. Backend returns a 30-day hashed token set in storage.
Already Authenticated Redirect	Navigate directly to <code>/login</code> while logged in	<code>GuestGuard</code> intercepts request (<code>hasSession === true</code>) and redirects instantly to <code>/dashboard</code> .

⚠ Validation Errors to Test

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE
Company Email	Leave blank & click Sign In	"Email is required"	Frontend Zod (<code>login.schema.ts</code>)
Company Email	Invalid email format (e.g. <code>admin@com</code>)	"Please enter a valid email"	Frontend Zod (<code>login.schema.ts</code>)
Password	Leave blank & click Sign In	"Password is required"	Frontend Zod (<code>login.schema.ts</code>)

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR
Invalid Email or Password	Submit incorrect password (1–4 failed attempts)	Toast notification displays: <code>"Invalid email or password. N attempt(s) remaining before lockout."</code> for 5 seconds.
Unverified Email (403)	Log in with account that hasn't completed email verification	Toast notification displays: <code>"Please verify your email address before logging in. Check your inbox for the verification link."</code> with an interactive Resend Email action button (10-second duration). Clicking action navigates to <code>/check-email</code> .
Deactivated Account (401)	Log in with user account marked <code>isActive: false</code> in DB	Toast notification displays: <code>"Your account has been deactivated. Please contact your administrator."</code> .
Lockout Rate Limiting (429)	5 consecutive failed login attempts	Backend returns <code>429 Too Many Requests</code> with <code>remainingSecs</code> (900s / 15 mins). Sign In button becomes disabled (<code>disabled={true}</code>). Red text displays: <code>"Too many attempts. Try again in {secondsLeft} seconds."</code> along with a Reset Password? link. Live countdown timer runs via <code>useCountdown</code> .
Server Offline / Network Failure	Submit login form when backend server is unreachable	Toast notification displays Axios error message (<code>"Login failed"</code> or <code>"Something went wrong"</code>).

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS?	POST-LOGIN REDIRECT TARGET	CODE REFERENCE
Unauthenticated / Guest	✔ Yes	N/A (Can view and submit login form)	<code>AppRoutes / GuestGuard</code>
<code>ORG_ADMIN</code>	✗ No (Redirected if logged in)	Redirects to <code>/dashboard</code>	<code>LoginView.tsx:L68</code>
<code>HR / HR_ADMIN</code>	✗ No (Redirected if logged in)	Redirects to <code>/hr/dashboard</code>	<code>LoginView.tsx:L71</code>
<code>EMPLOYEE</code>	✗ No (Redirected if logged in)	Redirects to <code>/employee/dashboard</code>	<code>LoginView.tsx:L74</code>
<code>BRANCH_ADMIN</code>	✗ No (Redirected if logged in)	Fallback redirects to <code>/</code>	<code>LoginView.tsx:L77</code>
<code>LEADERSHIP</code>	✗ No (Redirected if logged in)	Fallback redirects to <code>/</code>	<code>LoginView.tsx:L77</code>
<code>MANAGER</code>	✗ No (Redirected if logged in)	Fallback redirects to <code>/</code>	<code>LoginView.tsx:L77</code>
<code>PRODUCT_MANAGER</code>	✗ No (Redirected if logged in)	Fallback redirects to <code>/</code>	<code>LoginView.tsx:L77</code>

1.3 Password Recovery & Account Activation

- **Routes:** Password Recovery screens (`/auth/forgot-password` , `/auth/reset-password` , `/auth/verify-email`)
- **Testing Steps:**
 1. On the login screen, click the **Forgot Password?** link located below the password field (`/auth/forgot-password`).
 2. Enter your registered email address and submit.
- **⚠ Validation Errors to Test:**
 - Leaving email blank triggers `"Email is required"` .
 - Mismatched new password in Reset Password triggers `"Passwords do not match"` .

2. 📊 Main Dashboard & Quick Widgets

- **Route:** Main Dashboard screen (`/dashboard`)
- **Overview:** Centralized command center providing real-time workforce metrics, AI-driven insights, interactive attendance charts, pending leave requests, team widgets, and organization setup guidance.
- **Access Control:** Authenticated users only (`AuthGuard`). Redirects unauthenticated guests to `/login` .
- **Testing Steps:**
 1. Log in to the application. You will land directly on the **Main Dashboard** screen (`/dashboard`). Alternatively, click **Dashboard** in the top section of the left sidebar.
 2. Observe top header greeting displaying user's first name, last login timestamp, IP address, and login device.
 3. Review top KPI cards (*Total Employees*, *Present Today*, *Leave Requests Pending*, *Upcoming Celebrations*).
 4. (If logged in as `ORG_ADMIN` or `HR_ADMIN` with an incomplete workspace setup) Observe the **Organization Initial Setup Required** banner displaying current progress (0–3 steps completed). Click **Complete Initial Setup** to open the

modal wizard.

5. Fill out setup modal fields (**Country Code**, **Timezone**, **Base Currency**, **Fiscal Year Start**, **Employee Count Range**, **Industry**, **Phone Number**, **Admin Job Title**) and click **Complete Setup & Seed Head Office**.
6. (If logged in as non- `ORG_ADMIN`) Interact with the **Daily Punch Card** widget to clock in/out.

2.1 Organization Setup Wizard Dialog

 **How to Get Here:** After logging in, click **Dashboard** in the top section of the left sidebar (icon: `<DashboardIcon />`). This takes you to the **Main Dashboard** screen (`/dashboard`). If you are logged in as an administrator (`ORG_ADMIN` or `HR_ADMIN`) and your organization setup is incomplete (i.e. Head Office branch, departments, or designations have not yet been created), the purple **Organization Initial Setup Required**  banner will appear near the top of the dashboard page. Click **Complete Initial Setup** on the right side of this banner to open the setup wizard modal dialog (`AdminSetupWizardDialog`). If you do not see this banner, your organization structure is already fully configured or your role does not have setup permissions — see the Role-Based Access table below.

- **Route:** `/dashboard` (Modal overlay: `AdminSetupWizardDialog`)
- **Overview:** Interactive setup wizard allowing company administrators to configure organization locales (country, timezone, currency, fiscal year) and automatically seed Head Office branch, default departments, designations, shifts, and statutory national holidays into the database.
- **Testing Steps:**
 1. Log in as `ORG_ADMIN` or `HR_ADMIN` . On the Main Dashboard screen (`/dashboard`), locate the setup banner.
 2. Locate the **Organization Initial Setup Required**  card and click **Complete Initial Setup**.
 3. Verify the setup modal opens with pre-filled default locale values fetched from your tenant organization.
 4. Fill or adjust the 8 configuration fields (**Country Code**, **Timezone**, **Base Currency**, **Fiscal Year Start**, **Employee Count Range**, **Industry**, **Phone Number**, **Admin Job Title**).
 5. Click **Complete Setup & Seed Head Office**.
 6. Confirm the loading spinner runs, the modal closes, the success toast appears, and the setup banner on `/dashboard` auto-hides once all initial master data is seeded.

Step-by-Step Field Guide: Organization Setup Wizard

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Country Code	Select country from dropdown. • <i>Format:</i> 2-letter ISO code dropdown. • <i>Valid Example:</i> <code>India (IN)</code>	Yes	Pre-populated from tenant data (<code>orgData.countryCode</code>). Selecting a new country dynamically updates the flag icon and international dial code in the Phone Number field.
Timezone	Select primary timezone from dropdown. • <i>Format:</i> Standard IANA timezone string. • <i>Valid Example:</i> <code>Asia/Kolkata (IST +5:30)</code>	Yes	Pre-populated from <code>orgData.timezone</code> . Options: <code>Asia/Kolkata</code> , <code>UTC</code> , <code>America/New_York</code> , <code>Europe/London</code> , <code>Asia/Dubai</code> .

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Base Currency	Select base accounting currency. • <i>Format</i>: 3-letter currency code dropdown. • <i>Valid Example</i>: <code>INR</code> (₹) - Indian Rupee	Yes	Pre-populated from <code>orgData.baseCurrency</code> . Options: <code>USD</code> , <code>EUR</code> , <code>GBP</code> , <code>AED</code> .
Fiscal Year Start	Select fiscal year starting month. • <i>Format</i>: Month name dropdown. • <i>Valid Example</i>: <code>April</code>	Yes	Pre-populated from <code>orgData.fiscalYearStart</code> . Options: <code>April</code> , <code>January</code> , <code>July</code> , <code>October</code> .
Employee Count Range	Select estimated workforce scale. • <i>Format</i>: Range bracket dropdown. • <i>Valid Example</i>: <code>11-50</code>	Yes	Pre-populated from <code>orgData.employeeCountRange</code> . Options: <code>1-10 Employees</code> , <code>11-50 Employees</code> , <code>51-200 Employees</code> , <code>201-500 Employees</code> , <code>500+ Employees</code> . Automatically sets <code>maxEmployees</code> subscription limit on backend.
Industry	Select primary business sector. • <i>Format</i>: Category dropdown. • <i>Valid Example</i>: <code>Technology</code>	Yes	Pre-populated from <code>orgData.industry</code> . Options: <code>Technology</code> , <code>Healthcare</code> , <code>Financial Services</code> , <code>Retail & E-commerce</code> , <code>Manufacturing</code> , <code>Education</code> , <code>Professional Services</code> , <code>Other</code> .
Phone Number	Enter contact phone number. • <i>Format</i>: Exactly 10 numeric digits (digits only, no spaces or symbols). • <i>Valid Example</i>: <code>9876543210</code>	Yes	Uses <code><PhoneInput /></code> . Pre-filled from <code>orgData.phone</code> (stripped of non-digits). Automatically strips non-numeric characters as you type. Dial code prefix is linked to Country Code .
Admin Job Title	Enter administrator's official title. • <i>Format</i>: Free-text string (1-100 chars). • <i>Valid Example</i>: <code>Head of HR & Operations</code>	Yes	Plain text input. Must not be empty or whitespace-only. Saved to administrator profile.

Modal Action Buttons & Controls

- **Cancel Button**: Closes the dialog (`onClose()`) without saving changes or seeding database tables. Disabled while form submission is in progress (`submitting: true`).
- **Close (X) Icon**: Located at top-right of dialog header. Immediately dismisses the modal without submitting.
- **Complete Setup & Seed Head Office Button**: Triggers form submission (`handleSubmit`).
 - *Input State*: Requires **Phone Number** (10 digits) and **Admin Job Title** to be non-empty.
 - *Submitting State*: Disables both buttons, shows `<CircularProgress size={18} />`, and changes label to `"Completing & Seeding..."`.

Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Open Setup Wizard Dialog	Click Complete Initial Setup button on <code>OrgSetupGuidanceWidget</code> card	<code>AdminSetupWizardDialog</code> opens (<code>open={true}</code>). Backdrop blurs (<code>backdropFilter: "blur(6px)"</code>). Form controls pre-fill with tenant data from <code>useUserOrgData()</code> .	<code>OrgSetupGuidanceWidget.tsx:L169</code>
Country Flag & Phone Dial Sync	Change Country Code dropdown selection	Country flag icon and phone prefix inside <code><PhoneInput /></code> update dynamically to match the selected country code.	<code>AdminSetupWizardDialog.tsx:L337-L344</code>
Execute Setup & Master Data Seeding	Fill valid inputs & click Complete Setup & Seed Head Office	API POST <code>/api/v1/auth/complete-onboarding</code> returns HTTP 200. Backend creates <code>Head Office</code> branch (code <code>HQ</code>), seeds default roles, departments, designations, shifts, and national holidays. Toast displays <code>"Workspace configured successfully."</code> . Dialog closes and <code>fetchCounts()</code> triggers.	<code>AdminSetupWizardDialog.tsx:L126-L134</code> , <code>auth.service.ts:L751-L830</code>
Refresh Progress Counts	Click Refresh status icon () on <code>OrgSetupGuidanceWidget</code> header	<code>fetchCounts()</code> re-queries <code>getHeadOffice()</code> , <code>listDepartments(1,1)</code> , and <code>listDesignations(1,1)</code> . Progress bar updates to match completed steps (<code>0%</code> , <code>33%</code> , <code>66%</code> , <code>100%</code>).	<code>OrgSetupGuidanceWidget.tsx:L41-L69</code>
Auto-Hide Guidance Banner	Complete setup so Head Office, Departments, and Designations exist	<code>OrgSetupGuidanceWidget</code> checks <code>hasBranch && hasDepts && hasDesigns === true</code> and cleanly unmounts (<code>return null</code>) from <code>/dashboard</code> .	<code>OrgSetupGuidanceWidget.tsx:L96</code>

Validation Errors to Test

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
Phone Number	Enter a phone number with length <code>\$\neq 10\$</code> digits (e.g. <code>98765</code>) and submit	<code>"Contact phone number must be exactly 10 digits."</code>	Frontend <code>AdminSetupWizardDialog.tsx:L106-L109</code>
Phone Number	Submit with phone field completely empty	Form submission blocked by <code>!phone.trim()</code> check / HTML5 <code>required</code> attribute validation	Frontend <code>AdminSetupWizardDialog.tsx:L104</code>
Admin Job Title	Leave blank or enter whitespace only and submit	Form submission blocked silently by <code>!adminJobTitle.trim()</code> check / HTML5 <code>required</code> attribute validation	Frontend <code>AdminSetupWizardDialog.tsx:L104</code>

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
Country Code	Send single character or invalid code via direct API payload	"Country code must be 2 characters"	Backend Zod DTO common.validator.ts:L218
Timezone	Send empty string for timezone via direct API payload	"Timezone is required"	Backend Zod DTO auth.dto.ts:L92
Industry	Send string shorter than 2 characters via direct API payload	"Must be at least 2 character"	Backend Zod DTO common.validator.ts:L201
Admin Job Title	Send whitespace-only string " " via direct API payload	"Job title cannot be empty"	Backend Zod DTO auth.dto.ts:L104

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Missing Tenant Session (401)	Submit setup form without active JWT auth session (!req.context?.tenantId)	Backend returns HTTP 401. Dialog catches error and displays red alert: "Unauthorized".	auth.controller.ts:L227-L230
Organization Not Found (404)	Submit setup for a deleted/invalid tenantId	Backend returns HTTP 404. Dialog catches error and displays red alert: "Organization not found".	auth.service.ts:L753
API / Server Setup Failure (500)	Database transaction or seeding script fails during submission	Red <Alert severity="error"> banner renders inside dialog displaying err?.response?.data?.message or "Failed to complete onboarding setup.". Dialog remains open with entered values preserved.	AdminSetupWizardDialog.tsx:L137-L140
Head Office Check API Failure	getHeadOffice() API call fails on widget mount	OrgSetupGuidanceWidget silently catches error, sets hasBranch = false, allowing manual branch creation step to remain visible.	OrgSetupGuidanceWidget.tsx:L50-L52
Department / Designation Count API Failure	listDepartments() or listDesignations() API network failure	Promise.all catch block catches error, sets counts to 0, and displays top error alert: "Failed to check organization setup status".	OrgSetupGuidanceWidget.tsx:L63-L66

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS SETUP BANNER & DIALOG?	COMPONENT VISIBILITY / RESTRICTIONS	CODE REFERENCE
ORG_ADMIN	✓ Yes	Full access. <code>OrgSetupGuidanceWidget</code> renders on <code>/dashboard</code> . Can open setup modal, execute setup, and seed Head Office master data.	<code>OrgSetupGuidanceWidget.tsx:L72</code>
HR_ADMIN	✓ Yes	Full access. <code>OrgSetupGuidanceWidget</code> renders on <code>/dashboard</code> . Can open setup modal, execute setup, and seed Head Office master data.	<code>OrgSetupGuidanceWidget.tsx:L72</code>
EMPLOYEE	✗ No	Access restricted. <code>OrgSetupGuidanceWidget</code> returns <code>null</code> (guidance banner and setup wizard are completely hidden).	<code>OrgSetupGuidanceWidget.tsx:L87</code>
MANAGER	✗ No	Access restricted. <code>OrgSetupGuidanceWidget</code> returns <code>null</code> (guidance banner and setup wizard are completely hidden).	<code>OrgSetupGuidanceWidget.tsx:L87</code>
LEADERSHIP	✗ No	Access restricted. <code>OrgSetupGuidanceWidget</code> returns <code>null</code> (guidance banner and setup wizard are completely hidden).	<code>OrgSetupGuidanceWidget.tsx:L87</code>
BRANCH_ADMIN	✗ No	Access restricted. <code>OrgSetupGuidanceWidget</code> returns <code>null</code> (guidance banner and setup wizard are completely hidden).	<code>OrgSetupGuidanceWidget.tsx:L87</code>
PRODUCT_MANAGER	✗ No	Access restricted. <code>OrgSetupGuidanceWidget</code> returns <code>null</code> (guidance banner and setup wizard are completely hidden).	<code>OrgSetupGuidanceWidget.tsx:L87</code>

Step 3 — Flag List (Uncertainties & Rule Mismatches)

1. Frontend vs Backend Validation Discrepancies:

o Phone Number Length Rule:

- **Frontend** (`AdminSetupWizardDialog.tsx:L106`) strictly enforces `phone.replace(/\D/g, "").length === 10` and shows `"Contact phone number must be exactly 10 digits."`.
- **Backend DTO** (`auth.dto.ts:L97`) uses `optionalString(phoneSchema)`, which checks `min(1, "Phone is required")` and regex `/^\d+$/` ("Phone must contain only digits"), but does **not** enforce 10 digits. (Note: `withPhoneValidation` is defined in `common.validator.ts` but is not chained onto `OnboardingWizardDto`).

o Admin Job Title Empty State:

- **Frontend** (`AdminSetupWizardDialog.tsx:L104`) checks `!adminJobTitle.trim()` and returns silently without setting a state error string (relying on HTML5 `required` browser validation).
- **Backend DTO** (`auth.dto.ts:L104`) uses `optionalString(z.string().min(1, "Job title cannot be empty"))`. If sent as `""`, `optionalString` converts it to `undefined` (making it optional), but if sent as whitespace `" "`, Zod triggers `"Job title cannot be empty"`.

- **Fiscal Year Month Options:**

- **Frontend** UI dropdown only offers 4 months: `["April", "January", "July", "October"]`.
- **Backend DTO** (`auth.dto.ts:L99-L103`) accepts an enum of all 12 calendar months (`"January"` through `"December"`).

2. Toast Message Resolution:

- On success, `AdminSetupWizardDialog.tsx:L128-L131` attempts fallback `"Organization setup & Head Office seeding completed successfully!"`, but the backend API (`auth.service.ts:L826`) returns `message: "Workspace configured successfully."`. Therefore, `showSnackbar` displays the backend string (`"Workspace configured successfully."`).

3. Role Gating Consistency:

- `OrgSetupGuidanceWidget.tsx` explicitly permits `ORG_ADMIN` and `HR_ADMIN`. `BRANCH_ADMIN` is excluded from initial setup guidance because the wizard creates the root `Head Office` branch.

3. Employee Directory & People Hub

- **Routes:** Employee Directory (`/employees/directory`), Add Employee (`/employees/create`), Profile View (`/employees/profile/:id`)
- **Overview:** Complete workforce management hub with dual view modes (Table & Card Grid).

3.1 Directory View & Filters

- **Route:** Employee Directory screen (`/employees/directory`)  **How to Get Here:** Click **People** in the left sidebar (under top navigation group, icon: `<PeopleIcon />`). This opens the **Employee Directory** screen (`/employees/directory`).
- **Features & Testing:**
 1. **View Mode Switcher:**
 - Click **Grid View** icon to display employee cards with profile avatars.
 - Click **Table View** icon to display the high-performance **Virtualized Table**.
 2. **Search & Multi-Filtering:**
 - Type any employee name or code in the **Search** input.
 - Click the **Filter Bar** to filter by Department, Designation, or Status.

3.2 Add New Employee Form

 **How to Get Here:** After logging in: 1. Click **People** in the left sidebar (under top navigation group, icon: `<PeopleIcon />`). This opens the **Employee Directory** screen (`/employees/directory`). 2. Click the `[ + Add Employee ]` button in the top-right corner of the screen. This opens the **Add New Employee** form (`/employees/create`). If you do not see the **Employees** menu item or the **Add Employee** button, your role does not have `employee.read` or `employee.create` permissions — see the Role-Based Access table below.

- **Route:** `/employees/create` (wrapped in `<RoleGuard permission="employee.create">`)
- **Overview:** Comprehensive multi-card employee onboarding form featuring 7 structured cards, cascading organizational placement (Branch → Department → Designation → Eligible Managers), international phone formatting, bank account setup, and automatic salary structure mapping.
- **Testing Steps:**
 1. Log in as `ORG_ADMIN` or `HR_ADMIN`, click **People** in the left sidebar (`/employees/directory`), then click `[ + Add Employee ]` (`/employees/create`).

2. Complete **Card 1: Organizational Placement**: Select **Branch Location** (STEP 1). Observe that **Department** (STEP 2) unlocks. Select a Department; observe that **Designation** (STEP 3) and **Primary Reporting Manager** (STEP 5) populate with branch-and-department-filtered options. Select Designation and System Security Role (STEP 7).
3. Complete **Card 2: Basic Information & Contact**: Enter **First Name**, **Last Name**, **Work Email Address**, and **Mobile Number**.
4. Complete **Card 3: Employment Details & Schedule**: Select **Employee Type** (defaults to `Full-Time`) and **Joining Date** (`YYYY-MM-DD`).
5. (Optional) Complete **Card 4: Bank Account Details**: Enter **Bank Name**, **Account Number**, **IFSC Code**, and select **Account Type**.
6. (Optional) Complete **Card 5: Compensation & Salary Setup**: Enter **Annual CTC (₹)** and review or adjust the component breakdown (**Basic Salary**, **HRA**, **Special Allowance**).
7. Click  **Submit Employee**.
8. Confirm the loading spinner runs inside the button (`submitting: true`), the form submits to POST `/api/v1/employees`, a success toast is dispatched, and you are redirected to `/employees/directory`.

Step-by-Step Field Guide: Add New Employee Form

Card 1: Organizational Placement & Hierarchy

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
STEP 1: Branch Location	Select physical office branch. • <i>Format</i>: Select dropdown. • <i>Valid Example</i>: <code>Head Office</code>	Yes	Pre-populated with active branch from <code>useActiveBranchId()</code> . Changing Branch resets Department, Designation, and Manager fields, and triggers <code>listDepartments(1, 100, branchId)</code> .
STEP 2: Department	Select organizational department. • <i>Format</i>: Cascading dropdown. • <i>Valid Example</i>: <code>Engineering (ENG)</code>	Yes	Disabled until Branch Location is selected (<code>disabledPlaceholder: "Select Branch first"</code>). Changing Department resets Designation and Manager fields, and triggers <code>listDesignations(1, 100, departmentId)</code> .
STEP 3: Designation	Select official job designation. • <i>Format</i>: Cascading dropdown. • <i>Valid Example</i>: <code>Senior Full Stack Developer</code>	Yes	Disabled until Department is selected (<code>disabledPlaceholder: "Select Department first"</code>). Options are filtered by selected Department.
STEP 4: Squad Team	Select functional project squad/team. • <i>Format</i>: Select dropdown. • <i>Valid Example</i>: <code>Core Platform Squad (SQUAD)</code>	Optional	Populated from <code>listTeams()</code> . Optional placement tag.

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
STEP 5: Primary Reporting Manager	Select direct reporting manager. • <i>Format:</i> Cascading dropdown with sub-labels. • <i>Valid Example:</i> Vikram Malhotra [EMP-004] • Engineering Lead	Optional	Disabled until both Branch and Department are selected (<code>disabledPlaceholder: "Select Branch & Dept first"</code>). Populated dynamically by <code>useEligibleManagers()</code> . Auto-selects department head if default manager exists.
STEP 6: Secondary Managers	Select matrix/project managers. • <i>Format:</i> Multi-select dropdown. • <i>Valid Example:</i> Ananya Sharma [EMP-012] • Product Manager	Optional	Disabled until Branch and Department are selected. Supports multiple manager selections.
STEP 7: System Security Role	Select system access role. • <i>Format:</i> Select dropdown. • <i>Valid Example:</i> Employee (EMPLOYEE)	Yes	Options loaded from <code>listRoles()</code> API with fallback defaults (<code>EMPLOYEE</code> , <code>MANAGER</code> , <code>TEAM_LEADER</code> , <code>HR_ADMIN</code> , <code>BRANCH_ADMIN</code> , <code>ORG_ADMIN</code>). Defaults to <code>EMPLOYEE</code> .

Card 2: Basic Information & Contact

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
First Name	Enter employee's first name. • <i>Format:</i> Text string (2–100 chars). • <i>Valid Example:</i> Rohan	Yes	Plain text input. <code>autocomplete="new-password"</code> enabled to prevent browser autofill.
Last Name	Enter employee's last name. • <i>Format:</i> Text string (2–100 chars). • <i>Valid Example:</i> Sharma	Yes	Plain text input. <code>autocomplete="new-password"</code> enabled.
Work Email Address	Enter official company email address. • <i>Format:</i> Valid email string. • <i>Valid Example:</i> rohan.sharma@apexglobal.io	Yes	Validated for email format. Serves as unique login identity across tenant.
Mobile Number	Select country code & enter phone number. • <i>Format:</i> International phone component. • <i>Valid Example:</i> +91 9876543210	Optional	Uses <code><PhoneInput /></code> . Max 10 digits for subscriber portion. Strips non-digits automatically.

Card 3: Employment Details & Schedule

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Employee Type	Select employment category. • <i>Format:</i> Select dropdown. • <i>Valid Example:</i> Full-Time	Yes	Options: FULL_TIME, PART_TIME, CONTRACT, INTERN, CONSULTANT, TEMPORARY, FREELANCE. Defaults to FULL_TIME. Automatically updates Salary Pay Type (FULL_TIME → CTC, PART_TIME → HOURLY_RATE, CONTRACT → FIXED_MONTHLY, INTERN → STIPEND).
Joining Date	Select official date of joining. • <i>Format:</i> Calendar date picker (YYYY-MM-DD). • <i>Valid Example:</i> 2026-09-15	Yes	Formats date string to YYYY-MM-DD on submission using formatToYYYYMMDD().
Probation End Date	Select end date of probation period. • <i>Format:</i> Calendar date picker (YYYY-MM-DD). • <i>Valid Example:</i> 2026-12-15	Optional	Date picker input.
Shift Selection	Select assigned working shift. • <i>Format:</i> Select dropdown. • <i>Valid Example:</i> General Morning Shift (09:00 - 18:00)	Optional	Populated from listShifts(). Defaults to organization's default shift if left unselected.

Card 4: Bank Account Details (Payroll Setup)

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Bank Name	Enter financial institution name. • <i>Format:</i> Text string. • <i>Valid Example:</i> HDFC Bank	Optional	Plain text input.
Account Number	Enter bank account number. • <i>Format:</i> 6 to 20 numeric digits. • <i>Valid Example:</i> 50100432109876	Optional	Restricted to numeric format up to 20 digits.
IFSC Code	Enter bank branch IFSC code. • <i>Format:</i> 11-char uppercase alphanumeric string. • <i>Valid Example:</i> HDFC0001234	Optional	Auto-converts input to uppercase. Validated against regex /^[A-Z]{4}0[A-Z0-9]{6}\$/.

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Account Type	Select bank account type. • <i>Format:</i> Select dropdown. • <i>Valid Example:</i> <code>SaLary Account</code>	Optional	Options: <code>SALARY</code> , <code>SAVINGS</code> , <code>CURRENT</code> . Defaults to <code>SALARY</code> .
Account Holder Name	Enter account holder name. • <i>Format:</i> Free text string. • <i>Valid Example:</i> <code>Rohan Sharma</code>	Optional	If left blank, automatically pre-fills with <code>\${firstName} \${lastName}</code> on form submit.

Card 5: Compensation & Salary Setup

LABEL AS SHOWN ON SCREEN	WHAT TO ENTER OR SELECT	REQUIRED?	DEPENDENCIES & DYNAMIC BEHAVIORS
Annual CTC (₹)	Enter total annual CTC amount. • <i>Format:</i> Number input. • <i>Valid Example:</i> <code>1200000</code>	Optional	Numeric input. Used to calculate default line-item breakdown if components are unallocated.
Currency	Select compensation currency. • <i>Format:</i> Select dropdown. • <i>Valid Example:</i> <code>INR (₹)</code>	Optional	Options: <code>INR</code> , <code>USD</code> , <code>EUR</code> , <code>GBP</code> .
Basic Salary	Enter annual Basic salary component. • <i>Format:</i> Number input. • <i>Valid Example:</i> <code>600000</code>	Optional	Maps to component code <code>BASIC</code> . Defaults to 50% of CTC if left blank.
HRA	Enter House Rent Allowance component. • <i>Format:</i> Number input. • <i>Valid Example:</i> <code>300000</code>	Optional	Maps to component code <code>HRA</code> .
Special Allowance	Enter Special Allowance component. • <i>Format:</i> Number input. • <i>Valid Example:</i> <code>300000</code>	Optional	Maps to component code <code>ALLOWANCE</code> .

Action Buttons & Form Controls

- **Cancel Button:** Located at bottom right of form. Navigates back to `/employees/directory` without saving.
- **Submit Employee Button:** Triggers `handleSubmit(onSubmit, onInvalidForm)`.
 - *Valid State:* Disables button, renders `<CircularProgress size={24} color="inherit" />`, and dispatches `createEmployeeRequest(payload)`.
 - *Invalid State:* Prevents submission, logs error object, and renders a warning banner at top: `"Validation Error: Please check "[fieldName]" ([errorDetails])"`.

Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Open Add Employee Form	Navigate to <code>/employees/create</code> or click Add Employee button	<code>EmployeeCreateView</code> opens (<code><RoleGuard permission="employee.create"></code>). Active branch pre-selects in STEP 1 . Roles and Teams load from API.	<code>EmployeeCreateView.tsx:L72-L173</code>
Cascading Branch & Dept Selection	Select Branch, then Department	<code>listDepartments(1, 100, branchId)</code> and <code>listDesignations(1, 100, departmentId)</code> execute. Cascading dropdowns populate. <code>useEligibleManagers()</code> fetches filtered manager list.	<code>EmployeeCreateView.tsx:L208-L277</code>
Auto-Select Default Manager	Select Department that has a designated head	<code>useEligibleManagers</code> hook returns <code>defaultManagerId</code> . Form automatically sets <code>managerId</code> field.	<code>EmployeeCreateView.tsx:L280-L284</code>
Submit Valid Employee Form	Fill required fields & click  Submit Employee	Form passes Zod schema validation. Payload dispatches to POST <code>/api/v1/employees</code> . Backend creates Employee record & User account, generates atomic employee code (<code>EMP-xxx</code>), and returns HTTP 201. Redux triggers <code>success: true</code> , toast displays, and page redirects to <code>/employees/directory</code> .	<code>EmployeeCreateView.tsx:L340-L414</code> , <code>employee.service.ts:L67-L180</code>
Account Holder Name Auto-Fill	Fill Bank Account Number & IFSC, leave Account Holder Name blank	On submit, <code>onSubmit</code> handler automatically constructs <code>accountHolderName = "\${firstName} \${lastName}"</code> .	<code>EmployeeCreateView.tsx:L377-L379</code>

Validation Errors to Test

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
First Name	Enter < 2 characters or leave empty	<code>"First name must be 2-100 characters"</code>	Frontend <code>create-employee.schema.ts:L86</code>
Last Name	Enter < 2 characters or leave empty	<code>"Last name must be 2-100 characters"</code>	Frontend <code>create-employee.schema.ts:L87</code>
Work Email Address	Leave empty or enter invalid email format (e.g. <code>rohan@</code>)	<code>"Email is required"</code> / <code>"Please enter a valid email"</code>	Frontend <code>create-employee.schema.ts:L88</code>
Mobile Number	Enter > 10 digits (e.g. <code>98765432101</code>)	<code>"Phone number cannot exceed 10 digits"</code>	Frontend <code>create-employee.schema.ts:L89</code>
STEP 1: Branch Location	Submit form without branch selected	<code>"Branch is required"</code>	Frontend <code>create-employee.schema.ts:L100</code>
STEP 2: Department	Submit form without department selected	<code>"Department is required"</code>	Frontend <code>create-employee.schema.ts:L101</code>
STEP 3: Designation	Submit form without designation selected	<code>"Designation is required"</code>	Frontend <code>create-employee.schema.ts:L102</code>

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
STEP 7: System Security Role	Submit form without role selected	"Role is required"	Frontend <code>create-employee.schema.ts:L107</code>
Joining Date	Leave joining date empty	"Joining date is required"	Frontend <code>create-employee.schema.ts:L109</code>
Account Number	Enter invalid account number (< 6 digits or non-numeric)	"Account number must be 6-20 numeric digits"	Frontend <code>create-employee.schema.ts:L67</code>
IFSC Code	Enter invalid IFSC format (e.g. <code>HDFC123</code>)	"Invalid IFSC code format (e.g. HDFC0001234)"	Frontend <code>create-employee.schema.ts:L77</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Workspace Tier User Limit Reached (403)	Submit employee creation when tenant has reached <code>maxEmployees</code> capacity	API returns HTTP 403. Red alert banner displays: "User limit reached for your workspace tier (Team size range: X, Max allowed: Y users). You cannot add user/employee #Z. Please upgrade your workspace tier." .	<code>employee.service.ts:L83-L88</code>
Strict Branch Exclusivity Rule Violation (409)	Create an employee with an email/identity already actively assigned to another branch	API returns HTTP 409. Red alert banner displays: "Strict Branch Exclusivity Rule: This employee is already actively assigned to branch \"[Branch Name]\". An employee cannot belong to multiple branches simultaneously..." .	<code>employee.service.ts:L103-L109</code>
Duplicate Email Address (409)	Submit form with an email that already exists in the organization	API returns HTTP 409. Red alert banner displays: "Employee with email \"[email]\" already exists" or "A user account with email \"[email]\" already exists" .	<code>employee.service.ts:L113-L129</code>
Inactive / Invalid Branch (400)	Select a branch that has been deactivated or deleted in DB	API returns HTTP 400. Red alert banner displays: "Specified branch does not exist or is inactive in this organization." .	<code>employee.service.ts:L163-L165</code>
Unauthorized Session (401)	Submit form with an expired JWT access token	API returns HTTP 401. User is redirected to <code>/login</code> .	<code>auth.middleware.ts:L35</code>

🛡️ Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS ADD EMPLOYEE FORM?	COMPONENT VISIBILITY & BEHAVIOR	CODE REFERENCE
<code>ORG_ADMIN</code>	✅ Yes	Full access. Has <code>employee.create</code> permission. Can open <code>/employees/create</code> and submit new employee profiles.	<code>routes/index.tsx:L117-L122</code>

ROLE SLUG	CAN ACCESS ADD EMPLOYEE FORM?	COMPONENT VISIBILITY & BEHAVIOR	CODE REFERENCE
HR_ADMIN	✔ Yes	Full access. Has <code>employee.create</code> permission. Can open <code>/employees/create</code> and submit new employee profiles.	<code>routes/index.tsx:L117-L122</code>
EMPLOYEE	✘ No	Access restricted. Does not have <code>employee.create</code> . <code>RoleGuard</code> blocks route <code>/employees/create</code> and redirects to <code>/unauthorized</code> .	<code>RoleGuard.tsx:L28</code>
MANAGER	✘ No	Access restricted. Does not have <code>employee.create</code> . <code>RoleGuard</code> blocks route <code>/employees/create</code> and redirects to <code>/unauthorized</code> .	<code>RoleGuard.tsx:L28</code>
LEADERSHIP	✘ No	Access restricted. Does not have <code>employee.create</code> . <code>RoleGuard</code> blocks route <code>/employees/create</code> and redirects to <code>/unauthorized</code> .	<code>RoleGuard.tsx:L28</code>
BRANCH_ADMIN	✘ No	Access restricted. Does not have <code>employee.create</code> by default. <code>RoleGuard</code> blocks route <code>/employees/create</code> and redirects to <code>/unauthorized</code> .	<code>RoleGuard.tsx:L28</code>
PRODUCT_MANAGER	✘ No	Access restricted. Does not have <code>employee.create</code> . <code>RoleGuard</code> blocks route <code>/employees/create</code> and redirects to <code>/unauthorized</code> .	<code>RoleGuard.tsx:L28</code>

Step 3 — Flag List (Uncertainties & Rule Mismatches)

1. Frontend vs Backend Validation Discrepancies:

◦ Phone Number Regex Rule:

- **Frontend** (`create-employee.schema.ts:L89`) validates phone with regex `/^\d{1,10}$/` ("Phone number cannot exceed 10 digits").
- **Backend DTO** (`employee.dto.ts:L21` & `common.validator.ts:L34-L38`) uses `withPhoneValidation` and `phoneSchema` (`/^\d+$/`). It checks numeric digits, but does not limit length to 10 digits unless `withPhoneValidation` library parsing succeeds.

◦ Work Email Key Mapping:

- **Frontend** form field registers as `email`.
- **Backend DTO** (`employee.dto.ts:L143-L146`) accepts both `email` and `workEmail`, transforming `data.email` || `data.workEmail` into the final `email` property.

2. Role Fallback List:

- If the backend `listRoles()` API call fails or experiences network latency, `EmployeeCreateView.tsx:L53-L60` falls back to `DEFAULT_FALLBACK_ROLES` (`EMPLOYEE`, `MANAGER`, `TEAM_LEADER`, `HR_ADMIN`, `BRANCH_ADMIN`, `ORG_ADMIN`) to prevent blocking the form UI.

3. Salary Structure Pay Type Sync:

- Selecting `employeeType` automatically sets `salarySetup.employeePayType` and maps structure types (`FULL_TIME` → `CTC`, `PART_TIME` → `HOURLY_RATE`, `CONTRACT` / `TEMPORARY` / `FREELANCE` → `FIXED_MONTHLY`,

`INTERN` / `CONSULTANT` → `STIPEND`). If `manageSalary` is true and line items are empty, `onSubmit` automatically defaults `BASIC` component to 50% of annual CTC.

3.3 Employee Profile Tabs

- **Routes:** `/profile` (Self Profile), `/employees/:id` (Employee 360° Profile)
- **Overview:** Comprehensive employee profile hub providing a 360-degree view of employee personal details, organizational placement, reporting hierarchy, attendance logs, leave balances, bank accounts, salary structure, and uploaded verification documents. Features 6 active primary tabs (**Overview**, **Personal**, **Attendance**, **Leave**, **Payroll**, **Documents**), live profile picture cropping/uploading, emergency contact management, leave application, bank account linking, and salary structure assignment.

How to Get Here:

- **Self Profile:** After logging in, click **My Profile** (with the `<PersonIcon />`) located at the bottom of the left sidebar navigation. This navigates directly to `/profile` (defaulting to `/profile?tab=overview`). Accessible to all logged-in users regardless of role.
- **Employee 360° Profile:** Click **Employees** in the left sidebar (under main navigation), select **Directory** or **Employee List** (`/employees/directory`), and click on any employee row/card. This opens `/employees/:id`. Accessing another employee's profile requires `employee.read` permission.

Step-by-Step Field Guide: Profile Actions & Modals (By Section Tab)

◆ A. Overview Section Tab (`?tab=overview`)

1. Upload Profile Picture (Avatar) Dialog

- **Trigger:** Hover over or click the blue edit pencil icon (`<EditOutlinedIcon />`) at the bottom-right of the avatar frame on the hero header card.
- **Modal:** `UploadAvatarDialog.tsx`
- **Fields & Controls:**
 - **Image File Input:** Click the circular upload zone or **Change Photo** link. Accepts `.png`, `.jpg`, `.jpeg`, `.webp` image files up to **5MB**. Selecting an invalid file type displays `"Invalid file type. Please select an image file (PNG, JPG, WEBP)."`. Exceeding 5MB displays `"File size exceeds 5MB limit. Please select a smaller image."`.
 - **Interactive Image Cropper:** Renders a circular cropping canvas (`CROP_SIZE = 220px`). Drag the photo within the frame to position.
 - **Zoom Controls:** Use the zoom slider, **Zoom In** (`+`), **Zoom Out** (`-`), mouse wheel scroll, or **Reset** (`↺`) button to scale the image between `1.0x` and `3.0x`.
 - **Submit Button (`Upload Picture`):** Generates a 400×400 PNG crop canvas and submits via `PATCH /api/v1/employees/me/avatar` (or `PATCH /api/v1/employees/:id/avatar`). On success, dispatches toast `"Profile picture updated successfully"` and updates the header avatar across the app.

2. Add Skill or Expertise Modal

- **Trigger:** Click **+ Add Skill** chip on the **Skills & Expertise** card inside the **Overview** tab (`?tab=overview`).
- **Modal:** Built-in `Dialog` in `OverviewTab.tsx`
- **Fields & Controls:**
 - **Skill Name:** Text input for technical or domain skill (e.g. `Docker`, `Python`, `TypeScript`).
 - **Submit Button (`Add Skill`):** Appends skill chip to local profile state and displays toast `"Skill added to profile"`.

◆ B. Personal Section Tab (`?tab=personal`)

3. 🛠 Edit Personal Details & Address Modal

- **Trigger:** Click **Edit Details** at the top right of the **Personal Information** card inside the **Personal** tab (`?tab=personal`).
- **Modal:** Built-in `Dialog` in `ProfileView.tsx`
- **Fields & Controls:**
 - **Phone Number:** Plain text input. Enter a 10-digit mobile number (e.g. `9876543210`). Optional.
 - **Date of Birth:** Native date picker (`YYYY-MM-DD`). E.g. `1995-08-15` . Optional.
 - **Gender:** Dropdown select with options `MALE` , `FEMALE` , `OTHER` . Optional.
 - **Country Code:** Text input for 2-letter ISO country code (e.g. `IN`). Defaults to `IN` .
 - **Address Line 1:** Text input for street/house address (e.g. `123 MG Road, Koramangala`). Optional.
 - **City:** Text input for city (e.g. `Bangalore`). Optional.
 - **State:** Text input for state (e.g. `Karnataka`). Optional.
 - **Pincode / Zip:** Text input for postal code (e.g. `560001`). Optional.
 - **Submit Button (`Save Changes`):** Submits payload to `PATCH /api/v1/employees/me` . Displays toast `"Personal details and address updated successfully"` .

4. 🚨 Add Emergency Contact Modal

- **Trigger:** Click **Add Contact** (`+`) at the top right of the **Emergency Contacts** card in the **Personal** tab (`?tab=personal`).
- **Modal:** `EmergencyContactDialog.tsx`
- **Fields & Controls:**
 - **Full Name** (`name`): Exact text input. Required. E.g. `Anita Nair` .
 - **Relationship** (`relationship`): Exact text input. Required. E.g. `Mother` , `Father` , `Spouse` .
 - **Phone Number** (`phone`): Numeric phone input. Auto-strips non-digits (`replace(/\\D/g, "")`) and caps at 10 digits. Required. E.g. `9876500001` .
 - **Submit Button (`Save Contact`):** Appends contact to `emergencyContacts` array and updates profile via `PATCH /api/v1/employees/me` . Displays toast `"Emergency contact added successfully"` .

◆ C. Attendance Section Tab (`?tab=attendance`)

5. 📅 Attendance Log & Calendar View

- **Trigger:** Click the **Attendance** tab (`?tab=attendance`).
- **Controls & Views:**
 - **Date Range Filters:** **From Date** (defaults to 1st of current month `YYYY-MM-01`) and **To Date** (defaults to current date `YYYY-MM-DD`).
 - **View Switcher:** Toggle between **List View** (structured table with columns Date, Status, Check In, Check Out, Worked Hours, Actions) and **Calendar View** (FullCalendar interactive grid color-coded by status: Green = Present, Orange = Late/Half Day, Red = Absent, Blue = On Leave).
 - **Monthly Summary Card:** Displays statistics for the selected period (Total Working Days, Days Present, Days Late, Days Absent, Leaves Taken, Total Worked Hours).
 - **Attendance Record Detail Modal:** Click any log row in List View or event block in Calendar View to open detailed punch logs, IP/location metadata, and regularization status.

6. ➕ Manual Attendance Entry Modal

- **Trigger:** Click **+ Manual Entry** button in **Attendance** tab (visible only if user holds `attendance.create` permission).
- **Modal:** `ManualAttendanceDialog.tsx`
- **Fields & Controls:**
 - **Date:** Date picker (`YYYY-MM-DD`). E.g. `2025-06-10` . Required.
 - **Punch In Time:** Time picker (`HH:mm`). E.g. `09:15 AM` . Required.

- **Punch Out Time:** Time picker (`HH:mm`). E.g. `06:30 PM` . Required.
- **Attendance Status:** Select dropdown with options `PRESENT` , `LATE` , `HALF_DAY` , `ABSENT` . Required.
- **Notes / Reason:** Text input for admin override justification. Optional.
- **Submit Button (`Save Log`):** Submits payload to `POST /api/v1/attendance/manual` . Displays toast `"Attendance record added successfully"` .

7. 📄 Regularize Attendance Request Modal

- **Trigger:** Click **Regularize** button on any attendance log entry row or click **+ Regularize Request** in the **Attendance** tab (`?tab=attendance`).
- **Modal:** `RegularizeRequestDialog.tsx`
- **Fields & Controls:**
 - **Target Attendance Date:** Pre-filled or date picker (`YYYY-MM-DD`). E.g. `2025-06-15` . Required.
 - **Requested Check-In Time:** Time picker (`HH:mm`). E.g. `09:00 AM` . Required.
 - **Requested Check-Out Time:** Time picker (`HH:mm`). E.g. `06:00 PM` . Required.
 - **Reason for Regularization:** Textarea input explaining reason (e.g. `Forgot to punch in due to client meeting`). Required.
 - **Submit Button (`Submit Request`):** Submits payload to `POST /api/v1/attendance/regularize` . Displays toast `"Regularization request submitted successfully"` .

◆ D. Leave Section Tab (`?tab=leave`)

8. 🌿 Leave Balances & Requests Overview

- **Trigger:** Click the **Leave** tab (`?tab=leave`).
- **Components:**
 - **Leave Balances Grid:** Displays available leave balances by policy type (e.g. *Casual Leave*, *Sick Leave*, *Earned Leave*) with available days count and total quota.
 - **My Leave Applications Table:** Displays submitted leave requests with columns `LEAVE TYPE` , `DATES` , `DAYS` , `REASON` , `STATUS` (`PENDING` , `APPROVED` , `REJECTED` , `CANCELLED`).

9. 📄 Apply for Leave Modal

- **Trigger:** Click **Apply Leave** (`+`) button at the top right of the **Leave** tab (`?tab=leave`).
- **Modal:** `ApplyLeaveDialog.tsx`
- **Fields & Controls:**
 - **Leave Type:** Select dropdown displaying available leave policies and balances (e.g. `Casual Leave (CL) – Balance: 10`). Required.
 - **From Date:** Native date picker (`YYYY-MM-DD`). E.g. `2025-07-10` . Required.
 - **From Session:** Select dropdown with options `Full Day` (`FULL_DAY`), `First Half` (`FIRST_HALF`), `Second Half` (`SECOND_HALF`). Defaults to `FULL_DAY` .
 - **To Date:** Native date picker (`YYYY-MM-DD`). E.g. `2025-07-12` . Required.
 - **To Session:** Select dropdown with options `Full Day` (`FULL_DAY`), `First Half` (`FIRST_HALF`), `Second Half` (`SECOND_HALF`). Defaults to `FULL_DAY` .
 - **Reason for Leave:** Textarea input explaining leave reason (minimum 5 characters). Required. Entering less than 5 characters displays red error text: `"Reason must be at least 5 characters long."` .
 - **Submit Button (`Submit`):** Submits payload to `POST /api/v1/leave/requests` . Displays toast `"Leave application submitted successfully!"` .

◆ E. Documents Section Tab (`?tab=documents`)

10. 📄 Upload Verification Document Modal

- **Trigger:** Click **Upload Document** at the top right of the **Documents** tab (`?tab=documents`).
- **Modal:** `DocumentsTab.tsx`
- **Fields & Controls:**
 - **Document Type:** Select dropdown with options: `PAN Card` (`PAN`), `Aadhaar Card` (`AADHAAR`), `Passport` (`PASSPORT`), `Driving License` (`DRIVING_LICENSE`), `Offer Letter` (`OFFER_LETTER`), `Resume` (`RESUME`), `Degree` (`DEGREE`), `Experience Letter` (`EXPERIENCE`), `Other` (`OTHER`). Defaults to `PAN` . Required.
 - **Select File:** File input accepting `.jpg` , `.jpeg` , `.pdf` formats. Required. Selecting unsupported file types displays red alert: `"Only JPG, JPEG, and PDF files are allowed."` .
 - **Submit Button** (`Upload`): Uploads multipart form data to `POST /api/v1/documents` . Displays toast `"Document uploaded – awaiting HR verification"` .

◆ F. Payroll Section Tab (`?tab=payroll`)

11. 🏠 Add Bank Account Modal

- **Trigger:** Click **Add Bank Account** at the top right of the **Bank Accounts** card inside the **Payroll** tab (`?tab=payroll`).
- **Modal:** Built-in `Dialog` in `PayrollTab.tsx`
- **Fields & Controls:**
 - **Bank Name:** Text input for official bank name (e.g. `State Bank of India`). Required.
 - **Account Number:** Text input for bank account number (8–20 digits, e.g. `1234567890`). Required.
 - **IFSC Code:** Text input for 11-character IFSC code (e.g. `SBIN0001234`). Required. Auto-converts input to uppercase.
 - **Account Type:** Select dropdown with options `Salary` (`SALARY`), `Savings` (`SAVINGS`), `Current` (`CURRENT`). Defaults to `SALARY` .
 - **Set as primary account:** Toggle switch (`isPrimary`). Defaults to `false` .
 - **Submit Button** (`Save`): Submits payload to `POST /api/v1/employees/me/bank-accounts` (or `POST /api/v1/employees/:id/bank-accounts`). Displays toast `"Bank account added successfully"` .

12. 💰 Assign / Revise Salary Structure Modal

- **Trigger:** Click **Assign Salary Structure** or **Revise Salary Structure** button in the **Payroll** tab (`?tab=payroll`). Available only to `ORG_ADMIN` , `HR_ADMIN` , `isSuperAdmin` , or users holding `payroll.create` , `payroll.run` , or `employee.update` permissions.
- **Modal:** `SalaryStructureDialog.tsx`
- **Fields & Controls:**
 - **Annual CTC (₹)** (`ctcAnnual`): Number input for annual compensation (e.g. `1200000`). Required. Typing automatically calculates:
 - **Calculated Gross Monthly:** $\text{\$}\{\text{Annual CTC}\} / 12$ (e.g. ₹1,00,000 / mo).
 - **Basic Salary (Monthly):** 50% of monthly gross (e.g. ₹50,000).
 - **HRA (Monthly):** 25% of monthly gross (e.g. ₹25,000).
 - **Special Allowance (Monthly):** Remaining balance (e.g. ₹25,000).
 - **Currency:** Select dropdown with options `INR (₹)` , `USD ($)` , `EUR (€)` , `GBP (£)` . Defaults to `INR` .
 - **Basic Salary (Monthly):** Override input for basic component.
 - **HRA (Monthly):** Override input for House Rent Allowance component.
 - **Special Allowance (Monthly):** Override input for Special Allowance component.
 - **Submit Button** (`Save Salary Structure`): Submits payload to `POST /api/v1/payroll/structures` . Displays toast `"Salary structure assigned successfully!"` .

✓ Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Tab Navigation & Query Sync	Click any profile tab (Overview , Personal , Attendance , Leave , Payroll , Documents)	URL updates with query parameter (e.g. <code>/profile?tab=leave</code>). Selected tab renders wrapped in <code><Suspense></code> fallback loader.	<code>ProfileView.tsx:L108-L111</code>
Upload Avatar	Select PNG image <code>\$\le 5\text{MB}\$</code> , adjust zoom slider to <code>1.2x</code> , click Upload Picture	API <code>PATCH /api/v1/employees/me/avatar</code> returns 200. Toast displays <code>"Profile picture updated successfully"</code> . Header avatar updates with cache-busting timestamp <code>?t=...</code> .	<code>ProfileView.tsx:L143-L153</code> , <code>UploadAvatarDialog.tsx:L99-L103</code>
Add Skill	Click + Add Skill , enter <code>Docker</code> , click Add Skill	Skill chip added to Overview card, toast displays <code>"Skill added to profile"</code> .	<code>OverviewTab.tsx:L521-L528</code>
Edit Personal Details	Click Edit Details , update phone to <code>9876543210</code> & city to <code>Bangalore</code> , click Save Changes	API <code>PATCH /api/v1/employees/me</code> returns 200. Modal closes, profile reloads, and toast displays <code>"Personal details and address updated successfully"</code> .	<code>ProfileView.tsx:L267-L271</code>
Add Emergency Contact	Click Add Contact , enter Name: <code>Anita Nair</code> , Relationship: <code>Mother</code> , Phone: <code>9876500001</code> , click Save Contact	API <code>PATCH /api/v1/employees/me</code> appends contact. Modal closes and toast displays <code>"Emergency contact added successfully"</code> .	<code>PersonalTab.tsx:L61-L65</code>
Attendance Date Filter & Switcher	Select date range & toggle to Calendar View	Attendance logs reload for range. FullCalendar renders color-coded event badges (Green/Orange/Red/Blue).	<code>AttendanceTab.tsx:L71-L115</code>
Manual Attendance Entry	Click + Manual Entry , pick date & times, click Save Log	API <code>POST /api/v1/attendance/manual</code> returns 200. Toast displays <code>"Attendance record added successfully"</code> . Log is saved to database.	<code>AttendanceTab.tsx:L58-L60</code>
Submit Attendance Regularization	Click Regularize on a log row, enter reason & times, click Submit Request	API <code>POST /api/v1/attendance/regularize</code> returns 200. Toast displays <code>"Regularization request submitted successfully"</code> . Request appears in Regularization Requests sub-tab as <code>PENDING</code> .	<code>AttendanceTab.tsx:L63-L65</code>
Apply for Leave	Click Apply Leave , select Casual Leave, pick dates, enter reason <code>\$\ge 5\$</code> chars, click Submit	API <code>POST /api/v1/leave/requests</code> creates request. Modal closes and toast displays <code>"Leave application submitted successfully!"</code> . Request appears in My Leave Applications table with status <code>PENDING</code> .	<code>LeaveTab.tsx:L67-L114</code> , <code>ApplyLeaveDialog.tsx:L56-L67</code>

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Upload Document	Select type <code>PAN Card</code> , choose PDF file, click Upload	API <code>POST /api/v1/documents</code> uploads file. Dialog closes, input resets, and toast displays <code>"Document uploaded – awaiting HR verification"</code> .	<code>DocumentsTab.tsx:L103-L114</code>
Add Bank Account	Click Add Bank Account , enter Bank: <code>SBI</code> , Account: <code>1234567890</code> , IFSC: <code>SBIN0001234</code> , click Save	API <code>POST /api/v1/employees/me/bank-accounts</code> returns 200. Dialog closes and toast displays <code>"Bank account added successfully"</code> .	<code>PayrollTab.tsx:L113-L130</code>
Assign Salary Structure	Click Assign Salary Structure , enter Annual CTC <code>1200000</code> , click Save Salary Structure	API <code>POST /api/v1/payroll/structures</code> saves structure with components (Basic: 6,00,000, HRA: 3,00,000, Special Allowance: 3,00,000). Toast displays <code>"Salary structure assigned successfully!"</code> .	<code>SalaryStructureDialog.tsx:L111-L123</code>

⚠ Validation Errors to Test

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
Avatar File Type	Select a non-image file (e.g. <code>document.txt</code> or <code>file.pdf</code>) in avatar upload dialog	<code>"Invalid file type. Please select an image file (PNG, JPG, WEBP)."</code>	Frontend <code>UploadAvatarDialog.tsx:L62</code>
Avatar File Size	Select an image file larger than 5MB	<code>"File size exceeds 5MB limit. Please select a smaller image."</code>	Frontend <code>UploadAvatarDialog.tsx:L66</code>
Leave Reason Length	Enter a leave reason shorter than 5 characters (e.g. <code>Sick</code>) in Apply Leave modal	<code>"Reason must be at least 5 characters long."</code>	Frontend <code>ApplyLeaveDialog.tsx:L189</code>
Leave Application Mandatories	Leave Leave Type, From Date, or To Date empty	Submit button remains disabled (<code>'disabled={!leaveTypeId}</code>	
Document File Extension	Select a <code>.docx</code> or <code>.png</code> file in Document Upload dialog	<code>"Only JPG, JPEG, and PDF files are allowed."</code>	Frontend <code>DocumentsTab.tsx:L86</code>
Emergency Contact Fields	Leave Name, Relationship, or Phone blank	Save Contact button remains disabled (<code>disabled={!isValid}</code>)	Frontend <code>EmergencyContactDialog.tsx:L124</code>
Regularization Reason	Submit regularization request without a reason	Submit button blocked by <code>!reason.trim()</code> check	Frontend <code>RegularizeRequestDialog.tsx</code>
Bank Account Mandatory Fields	Leave Bank Name, Account Number, or IFSC Code blank	Save button remains disabled (<code>'disabled={!bankName.trim()</code>	

FIELD	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	SOURCE / FILE REFERENCE
IFSC Code Format	Send invalid IFSC string (e.g. <code>12345</code>) via API payload	"Invalid IFSC code format (e.g. <code>HDFC0000123</code>)"	Backend Zod DTO <code>employee.dto.ts:L256</code>
Salary Annual CTC	Clear Annual CTC or enter <code>0</code> and click Save Salary Structure	"Please enter a valid Annual CTC."	Frontend <code>SalaryStructureDialog.tsx:L97</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Account Without Employee Link	User account lacks an associated employee record (<code>user.employeeId</code> is undefined) and attempts avatar upload	Red alert banner displays: "Your account does not have an employee profile linked. Please contact your administrator."	<code>ProfileView.tsx:L139-L167</code>
Unauthorized Profile Fetch (401)	JWT token expires while loading <code>/profile</code>	Axios interceptor handles redirect to <code>/auth/login</code> .	<code>axios.ts</code>
Employee Profile Not Found (404)	Navigate to <code>/employees/invalid-id</code> for a deleted employee ID	Profile fetch fails; page gracefully falls back to empty profile state without crashing.	<code>ProfileView.tsx:L248-L251</code>
Leave Application Failure (500)	Server responds with 500 or insufficient leave balance	Red alert banner displays inside Apply Leave dialog displaying server error message. Dialog remains open for correction.	<code>ApplyLeaveDialog.tsx:L105-L107</code>
Attendance Log Fetch Failure (500)	API <code>GET /attendance/my-history</code> fails	Red alert banner renders in Attendance tab: "Failed to load attendance records."	<code>AttendanceTab.tsx:L75</code>
Document Upload Server Failure (500)	Server responds with 500 during file upload	Red alert banner displays inside dialog: <code>res.message</code> or "Failed to upload document". Dialog stays open for retry.	<code>DocumentsTab.tsx:L106-L116</code>

🛡️ Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS PROFILE?	TAB & FEATURE RESTRICTIONS	CODE REFERENCE
<code>ORG_ADMIN</code>	✅ Yes	Full access to self profile (<code>/profile</code>) and all employee 360° profiles (<code>/employees/:id</code>). Can view/edit personal info, assign salary structures, upload avatars, apply/approve leaves, and manage documents.	<code>ProfileView.tsx:L91-L93</code> , <code>PayrollTab.tsx:L159-L160</code>

ROLE SLUG	CAN ACCESS PROFILE?	TAB & FEATURE RESTRICTIONS	CODE REFERENCE
HR_ADMIN	✔ Yes	Full access to self profile and all employee 360° profiles. Can view/edit details, manage bank accounts, apply/approve leaves, and assign/revise salary structures.	ProfileView.tsx:L91-L93 , PayrollTab.tsx:L159-L160
EMPLOYEE	✔ Self Only	Access to <code>/profile</code> (Self Profile). Access to view other employees' 360° profiles (<code>/employees/:id</code>) requires <code>employee.read</code> permission. When viewing other profiles, Edit Details , Add Emergency Contact , Upload Document , Add Bank Account , and Apply Leave buttons are hidden (<code>!isViewingOther</code>). Attendance tab is hidden when viewing other employees unless <code>attendance.read</code> or <code>attendance.create</code> permission is granted (<code>canViewAttendance</code>).	ProfileView.tsx:L92 , PersonalTab.tsx:L83 , DocumentsTab.tsx:L125 , PayrollTab.tsx:L271
MANAGER	✔ Self & Team	Full access to self profile. Can view team member 360° profiles if granted <code>employee.read</code> . Salary assignment button in Payroll tab is visible only if holding <code>payroll.create</code> , <code>payroll.run</code> , or <code>employee.update</code> permissions.	PayrollTab.tsx:L159-L160
LEADERSHIP	✔ Yes	Can view self profile and employee profiles. Action buttons to edit personal info, bank accounts, or documents are restricted when viewing other employees (<code>isViewingOther: true</code>).	PersonalTab.tsx:L83
BRANCH_ADMIN	✔ Yes	Full access to self profile and branch employee profiles.	ProfileView.tsx:L92
PRODUCT_MANAGER	✔ Yes	Access to self profile and employee profiles based on permissions.	ProfileView.tsx:L92

Step 3 — Flag List (Uncertainties & Rule Mismatches)

1. Frontend vs Backend Validation Discrepancies:

- **IFSC Code Regex Rule:**
 - **Frontend** `addBankAccount` form (`PayrollTab.tsx:L107`) checks `!ifscCode.trim()` (allowing any string format).
 - **Backend DTO** (`employee.dto.ts:L256`) strictly validates IFSC using regex `/^[A-Z]{4}0[A-Z0-9]{6}$/` (e.g. `SBIN0001234`). If a user inputs `SBI123` , the frontend accepts it, but the backend rejects it with `"Invalid IFSC code format (e.g. HDFC0000123)"` .
- **Document Allowed File Types:**
 - **Frontend** `handleFileSelect` (`DocumentsTab.tsx:L85`) allows `.jpg` , `.jpeg` , `.pdf` .
 - **Backend DTO** (`document.dto.ts`) validates MIME types (`image/jpeg` , `application/pdf`).

2. Placeholder / Coming Soon Features:

- **AI Insights Banner & Actions:** In `OverviewTab.tsx:L444-L476` and `ProfileView.tsx:L674-L705` , the **AI Summary**, **Generate Review**, **Generate Promotion Summary**, **Recommend Training**, **Schedule 1:1**, and **Send Recognition** buttons display a snackbar warning `"AI Action [...] is coming soon!"` .
- **Performance, Learning, Assets, Timeline, Notes, Activity Tabs:** In `ProfileView.tsx:L709-L718` , selecting these tabs renders a placeholder card stating `"[Tab Name] Section – Detailed [Tab Name] information for [User] is`

loaded into this section."

3. Attendance Tab Visibility Gating:

- o `ProfileView.tsx:L92` checks `canViewAttendance = !isViewingOther || hasPermission("attendance.read") || hasPermission("attendance.create")`. If an employee views another employee's profile without attendance permissions and tries to navigate to `?tab=attendance`, `useEffect` automatically redirects them back to `?tab=overview`.

4. 🕒 Attendance & Time Tracking

- Routes: `/attendance`, `/reports`, `/attendance/regularizations`

4.1 Real-Time Check-In / Check-Out (Daily Punch Card)

- Route: `/dashboard` (Note: Navigating to `/attendance` directly auto-redirects to `/profile?tab=attendance`)
- Source Component: `DailyPunchCard.tsx`
- API Calls: `GET /api/v1/attendance/me/today`, `POST /api/v1/attendance/me/punch/web`, `GET /api/v1/attendance/shifts`

🕒 **How to Get Here:** After logging in, click **Dashboard** in the left sidebar (under the main navigation group, icon: `<DashboardIcon />`). This takes you to `/dashboard`. The **Daily Attendance Punch Card** sits at the top of the main dashboard page. If your profile is incomplete, the card will display a profile completion lock screen instead — see the Role-Based Access & Lock Restrictions table below.

📄 Interactive Controls & Button-by-Button Field Guide

The Daily Punch Card is a dynamic, multi-state interactive control that guides the user through their daily work shift. Below is the field-by-field breakdown of every control, indicator, and button on the card:

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	STATE DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Live Digital Clock	Digital time display (<code>HH:MM:SS AM/PM TimeZone</code>), e.g. <code>09:30:15 AM IST</code>	Indicator	Continuously updates every 1 second (<code>1000ms</code>) using locale-aware formatting (<code>Intl.DateTimeFormat</code>). Displays current weekday, date, month, and year below the clock.
Geofence Verified Chip	Status badge (<code>#DCFCE7</code> background, green text)	Indicator	Displays green location icon (<code><LocationOnIcon /></code>) confirming client browser geolocation verification capability.
Shift Status Chip	Dynamic badge	Indicator	Displays real-time status: • <code>Not Clocked In</code> (Grey) — Before first check-in. • <code>Active Shift</code> (Green) — Punched in & actively working. • <code>On Break</code> (Orange) — Currently on break. • <code>Shift Completed</code> (Indigo) — Punched out for the day.

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	STATE DEPENDENCIES & INTERACTIVE UI BEHAVIOR
[Clock In] Button	Primary Action Button (Solid Indigo #6D5DF6)	Click Action	Visible when: User has not clocked in today. Behavior: Clicking prompts browser for HTML5 Geolocation coordinates (longitude , latitude), resolves default shift ID (/attendance/shifts), and submits type: "CHECK_IN" . Button shows spinner ("Clocking In..."). Result: Displays green alert "Checked in successfully!" , status chip changes to Active Shift , and card displays shift start timestamp. If profile completion is in NUDGE phase, pops open Nudge Reminder Modal.
Check-In Recorded Box	Info Box (Green dashed border)	Indicator	Visible when: Active shift is running. Content: Displays <CheckCircleOutlinedIcon /> and text: "Check-In Recorded — Shift started at HH:MM AM/PM today."
[Break Out] Button	Secondary Action Button (Orange #F59E0B)	Click Action	Visible when: User is in an Active Shift . Behavior: Submits type: "BREAK_OUT" . Button shows spinner ("Processing..."). Result: Displays green alert "Took a break successfully!" , status chip changes to On Break , card transitions to On Break info box.
[Resume Shift] Button	Action Button (Green #10B981)	Click Action	Visible when: User is currently On Break . Behavior: Submits type: "BREAK_IN" . Button shows spinner ("Resuming..."). Result: Displays green alert "Resumed shift successfully!" , status chip returns to Active Shift .
[Clock Out] Button	Danger Action Button (Red #EF4444)	Click Action	Visible when: User is in an Active Shift (side-by-side with Break Out). Behavior: Submits type: "CHECK_OUT" . Button shows spinner ("Clocking Out..."). Result: Displays green alert "Checked out successfully!" , status chip changes to Shift Completed , and card locks into Shift Summary mode.
Shift Summary Card	Summary Box (Indigo border)	Indicator	Visible when: Shift is completed (hasCheckedOut). Content: Displays: • Clock In: Start time + Late chip if isLate: true . • Clock Out: End time + Early chip if isCheckOutEarly: true . • Worked Hours: Computed duration (e.g. 8h 30m).
[Complete Profile] Button	Warning Action Button (Primary Indigo)	Click Action	Visible when: User profile is incomplete (isBlocked: true). Behavior: Clicking navigates directly to /onboarding . All punch buttons remain hidden until required profile sections are completed.

Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Initial Clock In	1. Open <code>/dashboard</code> 2. Confirm status chip reads <code>Not Clocked In</code> 3. Click <code>[Clock In]</code> .	Button shows spinner (<code>"Clocking In..."</code>). Green alert banner displays <code>"Checked in successfully!"</code> . Status chip updates to <code>Active Shift</code> . Check-In Recorded box displays <code>Shift started at [HH:MM AM/PM] today.</code>	<code>DailyPunchCard.tsx:L105-L110</code>
Start Break	1. During an active shift, click <code>[Break Out]</code> .	Button shows spinner (<code>"Processing..."</code>). Green alert banner displays <code>"Took a break successfully!"</code> . Status chip changes to <code>On Break</code> . Card renders <code>Currently On Break</code> box displaying break start time.	<code>DailyPunchCard.tsx:L155-L161</code>
Resume Shift	1. While on break, click <code>[Resume Shift]</code> .	Button shows spinner (<code>"Resuming..."</code>). Green alert banner displays <code>"Resumed shift successfully!"</code> . Status chip returns to <code>Active Shift</code> . Action buttons revert to <code>Break Out</code> and <code>Clock Out</code> .	<code>DailyPunchCard.tsx:L177-L183</code>
Clock Out	1. During an active shift, click <code>[Clock Out]</code> .	Button shows spinner (<code>"Clocking Out..."</code>). Green alert banner displays <code>"Checked out successfully!"</code> . Status chip changes to <code>Shift Completed</code> . Card renders Shift Summary with Clock In, Clock Out, and Worked Hours (<code>formatWorkedTime</code>).	<code>DailyPunchCard.tsx:L133-L137</code>
Nudge Modal Trigger	1. Clock in as an employee with profile in NUDGE phase.	Green alert appears AND Nudge Reminder Modal pops open displaying profile completion percentage (e.g. <code>45%</code>) with option to complete profile or dismiss.	<code>DailyPunchCard.tsx:L110-L114</code>

Validation Errors to Test

FIELD / DTO CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Punch Type Enum (<code>PunchDto</code>)	Send invalid punch type string to <code>POST /api/v1/attendance/me/punch/web</code>	Backend Zod schema validation fails: <code>"Invalid enum value. Expected 'CHECK_IN' 'BREAK_OUT' 'BREAK_IN' 'CHECK_OUT'"</code>	<code>attendance.dto.ts:L7</code>
Latitude Range (<code>PunchDto</code>)	Send latitude <code>< -90</code> or <code>> 90</code>	Backend Zod schema validation fails: <code>"Number must be greater than or equal to -90"</code> or <code>"Number must be less than or equal to 90"</code>	<code>attendance.dto.ts:L8</code>
Longitude Range (<code>PunchDto</code>)	Send longitude <code>< -180</code> or <code>> 180</code>	Backend Zod schema validation fails: <code>"Number must be greater than or equal to -180"</code> or <code>"Number must be less than or equal to 180"</code>	<code>attendance.dto.ts:L9</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Duplicate Clock In (400)	Attempting to clock in when already checked in	Red inline Alert banner displays server error message: <code>err.response.data.message</code> (e.g., "Already checked in for today"). Button spinner stops and error can be dismissed via <code>x</code> icon.	<code>DailyPunchCard.tsx:L119-L121</code>
Duplicate Clock Out (400)	Attempting to clock out when session is already closed	Red inline Alert banner displays server error message: <code>err.response.data.message</code> (e.g., "Already checked out for today").	<code>DailyPunchCard.tsx:L140-L142</code>
Shift Window Rejection (400)	Punching in before allowed shift start time when <code>rejectEarlyPunch: true</code>	Red inline Alert banner displays: "Early check-in not allowed for this shift" .	<code>attendance.service.ts:L142</code>
Hard Profile Lock (403)	User profile is incomplete (<code>PROFILE_INCOMPLETE_HARD</code>)	<code>useProfileBlockDetect</code> catches the 403 response (<code>isBlocked: true</code>). Punch card replaces clock buttons with a lock screen showing <code><LockOutlinedIcon /></code> , title "Profile Incomplete" , pending section chips, and a <code>[Complete Profile]</code> button.	<code>DailyPunchCard.tsx:L299-L372</code>
Network / Server Error (500)	Server responds with 500 or connection times out	Red inline Alert banner displays: "Something went wrong during check-in" (or <code>check-out</code> / <code>break-out</code> / <code>break-in</code>).	<code>DailyPunchCard.tsx:L120</code>
Geolocation Unsupported / Denied	Browser geolocation permission is blocked	<code>getLocation()</code> safely returns <code>null</code> . Punch request proceeds with <code>undefined</code> coordinates without crashing the UI.	<code>DailyPunchCard.tsx:L41-L48</code>

🛡 Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS PUNCH CARD?	ROLE-SPECIFIC BEHAVIOR & RESTRICTIONS	CODE REFERENCE
<code>ORG_ADMIN</code>	✅ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>
<code>HR_ADMIN</code>	✅ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>
<code>BRANCH_ADMIN</code>	✅ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>
<code>LEADERSHIP</code>	✅ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>

ROLE SLUG	CAN ACCESS PUNCH CARD?	ROLE-SPECIFIC BEHAVIOR & RESTRICTIONS	CODE REFERENCE
MANAGER	✓ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>
PRODUCT_MANAGER	✓ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches.	<code>DailyPunchCard.tsx:L28</code>
EMPLOYEE	✓ Yes	Can view daily punch card on <code>/dashboard</code> and record self attendance punches. Access is strictly self-service (stamped via <code>req.context.userId</code>). If profile is incomplete, profile completion lock activates.	<code>DailyPunchCard.tsx:L28</code> , <code>attendance.routes.ts:L37</code>

4.2 Attendance Reports & Punch Log Table

- **Route:** `/reports`
- **Source Component:** `AttendanceReportView.tsx`
- **API Calls:** `GET /api/v1/attendance/report` , `GET /api/v1/departments` , `GET /api/v1/designations` , `GET /api/v1/branches` , `GET /api/v1/attendance/shifts`
- **Permission Guard:** `report.read` (gated via `<RoleGuard permission="report.read">`)

🔗 **How to Get Here:** After logging in, click **Attendance Report** in the left sidebar (under the main navigation group, icon: `<AssessmentIcon />`). This takes you to `/reports` . Note: This page requires the `report.read` permission. If an Employee without `report.read` attempts to access this page, they are redirected or presented with their personal attendance history (`<AttendanceTab hideTabs={true} />`).

🔧 Interactive Controls, Filters & Component Guide

The Attendance Report page provides high-level workforce KPIs, trend analytics, multi-select filter bars, and a granular punch log data table. Below is the comprehensive field-by-field breakdown of every filter, visual chart component, and table element:

1. Filter Bar & Search Controls

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	FILTER DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Search Input	Text input (<code>searchPlaceholder="Search employees..."</code>), e.g. <code>EMP-001</code> or <code>John</code>	Optional	Typing filters the punch log table live by matching employee code or full name (<code>searchQuery</code>). Resets pagination to page 1 (<code>clientPage: 0</code>).
Date Range (<code>fromDate</code> / <code>toDate</code>)	Date range picker (<code>YYYY-MM-DD</code>), e.g. <code>2026-09-01</code> to <code>2026-09-30</code>	Optional	Defaults <code>fromDate</code> to 1st day of current year (<code>YYYY-01-01</code>) and <code>toDate</code> to today (<code>todayStr</code>). Changing dates refetches punch logs from <code>GET /api/v1/attendance/report</code> .
Branch Filter	Multi-select dropdown (<code>branchId</code>)	Optional	Populated dynamically from <code>GET /api/v1/branches</code> . Defaults to <code>All Branches</code> . Selecting specific branches filters punch logs by employee branch ID.

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	FILTER DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Departments Filter	Multi-select dropdown (<code>departmentId</code>)	Optional	Populated dynamically from <code>GET /api/v1/departments</code> . Defaults to <code>ALL Departments</code> . Multi-select dropdown filtering records by department.
Designation Filter	Multi-select dropdown (<code>designationId</code>)	Optional	Populated dynamically from <code>GET /api/v1/designations</code> . Defaults to <code>ALL Designations</code> . Multi-select dropdown filtering records by designation.
Status Filter	Multi-select dropdown (<code>status</code>)	Optional	Dropdown options: <code>Present</code> (<code>PRESENT</code>), <code>Absent</code> (<code>ABSENT</code>), <code>Late</code> (<code>LATE</code>), <code>Half Day</code> (<code>HALF_DAY</code>), <code>On Leave</code> (<code>ON_LEAVE</code>), <code>Holiday</code> (<code>HOLIDAY</code>), <code>Week Off</code> (<code>WEEK_OFF</code>). Defaults to <code>ALL Statuses</code> .
[Reset Filters] Button	Action Button	Click Action	Resets search query, clears date ranges, resets all dropdown filters to <code>ALL</code> , and resets table pagination to page 1.
[Mark Attendance] Button	Primary Action Button (Indigo <code>#6D5DF6</code>)	Click Action	Visible when: User holds <code>attendance.create</code> permission. Behavior: Clicking opens Manual Attendance Dialog (<code>ManualAttendanceDialog.tsx</code>) to manually insert an employee attendance log.

2. 📊 Weekly Attendance Trend Bar Chart (`WeeklyTrendBarChart.tsx`)

- **Header:** `"Weekly Attendance Trend"`
- **Y-Axis & Scale:** Renders dynamic Y-axis scale ticks (`0` , `25%` , `50%` , `75%` , `effectiveMax`) with dashed horizontal gridlines. Scale maximum is computed dynamically from max present count (`Math.max(...counts)`).
- **Bar Visuals:** Displays 7 vertical columns representing attendance counts for the last 7 days leading up to today (`Today` , `Sep 2` , `Sep 1` , etc.). Bars render in solid red (`#EF4444`) with rounded top corners.
- **Hover & Tooltip Behavior:** Hovering over a bar highlights it in darker red (`#DC2626`) with a scale effect (`transform: scaleY(1.03)`) and reveals an interactive tooltip:
 - **Present Days:** Displays `"[Date]: [Count] Present"` (e.g. `"Today: 32 Present"`).
 - **Weekend / Off Days:** Displays `"[Date]: 0 Present (Sunday - Week Off)"` or `"0 Present (Saturday - Week Off)"` .

3. 📈 Today's Status Breakdown & AI Insight Card (`TodayStatusBreakdownCard.tsx`)

- **Header:** Displays dynamic localized date title (e.g. `"Today - 03 Sep 2026"`).
- **Status Distribution Progress Bars:** Features 5 horizontal MUI `<LinearProgress>` bars showing headcount and percentage distribution for today:
 - **On Time:** Green bar (`#10B981`) showing on-time check-in count.
 - **Late:** Orange bar (`#F59E0B`) showing late check-in count (punched after shift start + grace period).
 - **WFH:** Primary Indigo bar (`#6D5DF6`) showing remote workforce count.
 - **Absent:** Red bar (`#EF4444`) showing absent count.
 - **On Leave:** Purple bar (`#8B5CF6`) showing employees on approved leave.
- **AI Insight Widget:** Displays AI spark icon (`<AutoAwesomeIcon />`), `"Coming Soon"` chip, and analytical trend snippet (`"Attendance dip on Fridays averages 8.2% above weekly average..."`). Includes `Explain` , `Save` , and `Dismiss` chips (`Dismiss` hides the widget).

4. Employee Attendance Table (`EmployeeAttendanceTable.tsx`)

- **Header & Export Action:** Card title `"Employee Attendance"` with right-aligned `[ Export ]` button (`<FileDownloadOutlinedIcon />`) that downloads current client-filtered rows as a CSV file (`Employee_Attendance_Today.csv`).
- **Virtualized Body & Performance:** Rendered using `<VirtualizedTableBody>` with `estimateRowHeight={56}` and `minWidth={650}` for smooth rendering of large punch datasets without vertical scrollbar clutter.
- **Columns:**
 1. `S.NO.` : Continuous 1-based row index calculated across pages (`startIndex + index + 1`).
 2. `EMPLOYEE CODE` : Displays employee code badge (e.g. `EMP-001`) or `--` .
 3. `EMPLOYEE NAME` : Displays employee circular avatar (`CustomAvatar`) + full name.
 4. `PUNCH LOG` : Formatted 12-hour timestamp (e.g. `09:15 AM`) or `--` .
 5. `PUNCH DATE` : Formatted date (e.g. `03 09 26`) or `--` .
- **Row Click Action:** Clicking any row highlights it (`#F9FAFB`) and opens the **Attendance Details Modal** displaying record status badge, check-in, check-out, and worked duration.
- **Pagination Controls:** MUI `<TablePagination>` footer providing `10` , `25` , `50` , `100` rows per page options. Default: `25` rows per page.

Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Load Attendance Report	1. Open <code>/reports</code> 2. Observe page header, KPI cards, and punch log table.	Page title displays <code>"Attendance"</code> with subtitle <code>"Real-time workforce attendance tracking"</code> . Top KPI cards calculate workforce Present Rate, Absent Count, Late Count, and WFH Count.	<code>AttendanceReportView.tsx:L593-L622</code>
Search Employee Punch Logs	1. Type <code>EMP-005</code> in search bar.	Punch log table filters live to show only rows matching <code>EMP-005</code> . Table page resets to 1.	<code>AttendanceReportView.tsx:L498-L530</code>
Filter by Status & Branch	1. Select Status: <code>Late</code> 2. Select Branch: <code>Headquarters</code> .	Table updates to show only <code>LATE</code> punch log entries registered under <code>Headquarters</code> branch.	<code>AttendanceReportView.tsx:L500-L527</code>
View Attendance Detail Modal	1. Click on any row in the Punch Log table.	Modal title displays <code>"Attendance Details"</code> . Renders Status Chip, First Check In, Last Check Out, and Worked Hours.	<code>AttendanceReportView.tsx:L750-L790</code>
Export CSV Report	1. Click <code>[Export]</code> button above table.	Browser downloads <code>Employee_Attendance_Today.csv</code> file containing <code>S.No</code> , <code>Employee Code</code> , <code>Employee Name</code> , <code>Punch Log</code> , and <code>Punch Date</code> .	<code>EmployeeAttendanceTable.tsx:L51-L77</code>
Change Rows Per Page	1. Select <code>50</code> in Table Pagination dropdown.	Table expands to display up to 50 rows per page.	<code>AttendanceReportView.tsx:L735-L745</code>

⚠ Validation Errors to Test

FIELD / QUERY CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Date Format (<code>AttendanceReportQueryDto</code>)	Send invalid date string (e.g. <code>01-09-2026</code>) in <code>fromDate</code> or <code>toDate</code> API call	Backend Zod schema validation fails: <code>"Date must be in YYYY-MM-DD format"</code>	<code>common.validator.ts:L94</code>
Branch ID Format (<code>AttendanceReportQueryDto</code>)	Send malformed MongoDB ID string in <code>branchId</code>	Backend Zod schema validation fails: <code>"Invalid ID format"</code>	<code>common.validator.ts:L7</code>
Page Size Limit (<code>AttendanceReportQueryDto</code>)	Request <code>pageSize</code> > 100 in API query	Backend Zod schema validation fails: <code>"Page size cannot exceed 100"</code>	<code>common.validator.ts:L24</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Unauthorized Access (403)	User without <code>report.read</code> navigates to <code>/reports</code>	Router <code><RoleGuard permission="report.read"></code> blocks access. <code>AttendanceReportView.tsx</code> renders fallback employee view <code><AttendanceTab hideTabs={true}></code> <code>/></code> titled <code>"My Attendance"</code> .	<code>AttendanceReportView.tsx:L577-L587</code>
Server Error on Report Fetch (500)	Backend DB connection failure during <code>GET /attendance/report</code>	Red inline Alert banner displays: <code>err.response.data.message</code> or <code>"Failed to load attendance report"</code> .	<code>AttendanceReportView.tsx:L189-L191</code>
Empty Table Results	Filter criteria returns 0 records	Punch Log table renders 0 rows with Table Pagination showing <code>0 of 0</code> .	<code>AttendanceReportView.tsx:L734</code>
Missing Employee Name Fallback	Attendance record has deleted or unlinked employee	Table displays employee name fallback <code>"Employee"</code> or <code>"N/A"</code> and employee code <code>--</code> without throwing a runtime <code>NullPointerException</code> .	<code>AttendanceReportView.tsx:L438</code>

🛡 Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS <code>/REPORTS</code> ?	ACCESS SCOPE & FEATURE RESTRICTIONS	CODE REFERENCE
<code>ORG_ADMIN</code>	✅ Full Access	Full access to <code>/reports</code> . Can view organization-wide attendance, filter by branch/dept/designation, export CSV, and click <code>[Mark Attendance]</code> (<code>canMarkAttendance: true</code>).	<code>AttendanceReportView.tsx:L111-L113</code>

ROLE SLUG	CAN ACCESS <code>/REPORTS</code> ?	ACCESS SCOPE & FEATURE RESTRICTIONS	CODE REFERENCE
<code>HR_ADMIN</code>	✅ Full Access	Full access to <code>/reports</code> . Can view organization-wide attendance, filter records, export CSV, and mark manual attendance (<code>canMarkAttendance: true</code>).	<code>AttendanceReportView.tsx:L113-L117</code>
<code>BRANCH_ADMIN</code>	✅ Branch Scope	Access to <code>/reports</code> . Branch filter options are restricted to their assigned branch (<code>canReadBranches: true</code>).	<code>AttendanceReportView.tsx:L116-L133</code>
<code>LEADERSHIP</code>	✅ Read-Only	Access to <code>/reports</code> . Can view attendance KPIs, trends, and punch table. <code>[Mark Attendance]</code> button is hidden (<code>canMarkAttendance: false</code>).	<code>AttendanceReportView.tsx:L598</code>
<code>MANAGER</code>	✅ Team Scope	Access to <code>/reports</code> (via <code>report.read</code>). Can view attendance punch logs for team members. Mark Attendance button is visible only if granted <code>attendance.create</code> .	<code>AttendanceReportView.tsx:L598</code>
<code>PRODUCT_MANAGER</code>	✅ Permission-Based	Access determined by <code>report.read</code> and <code>attendance.create</code> permissions.	<code>AttendanceReportView.tsx:L110-L117</code>
<code>EMPLOYEE</code>	❌ Restricted	Sidebar Navigation: "Attendance Report" sidebar item is hidden (<code>permission: "report.read"</code>). Direct Route Access: Navigating to <code>/reports</code> triggers <code>isEmployeeRole = true</code> fallback, rendering self attendance view (<code><AttendanceTab hideTabs={true} /></code>) titled "My Attendance". Cannot view organization-wide punch logs or mark manual attendance.	<code>DashboardLayout.tsx:L74</code> , <code>AttendanceReportView.tsx:L577-L587</code>

4.3 Attendance Regularization Requests & HR Approvals

- **Route:** `/attendance/regularizations`
- **Source Component:** `RegularizationListPage.tsx` , `ReviewRegularizationDialog.tsx` , `RegularizeRequestDialog.tsx`
- **API Calls:** `GET /api/v1/attendance/regularizations/pending` , `PATCH /api/v1/attendance/regularizations/:id/review` , `GET /api/v1/attendance/regularizations/me` , `POST /api/v1/attendance/regularizations`

 **How to Get Here:** After logging in as **HR Admin**, **Branch Admin**, or **Manager**, click **Regularization** in the left sidebar (under top navigation group, icon: `<CalendarMonthIcon />`). This opens `/attendance/regularizations` .

- **Sidebar Menu Permission:** Guarded by `permission: "leave.read"` .
- **HR / Admin / Manager View:** Renders the organization or branch pending regularization review queue (`getPendingRegularizationRequests`).
- **Employee View:** If accessed by an `EMPLOYEE` role, renders their personal regularization request history (`getMyRegularizationRequests`).

Step-by-Step HR Review Guide & Interactive Control Breakdown

As an HR Admin or Manager, the Regularization page allows you to inspect pending attendance correction requests submitted by employees, review their requested check-in/out adjustments and reasons, and approve or reject the request. Below is the field-by-field breakdown of the queue table and the review dialog:

1. HR Regularizations Queue Table (`RegularizationListPage.tsx`)

ELEMENT / COLUMN HEADER	FORMAT & TYPE	HR ACTION & INTERACTIVE UI BEHAVIOR
Page Header & <code>[Refresh]</code> Button	Header Bar	Displays calendar icon and title "Attendance Regularizations" . Clicking the top-right <code>[Refresh]</code> button manually refetches the latest pending request queue.
<code>EMPLOYEE</code>	Avatar + Name	Displays employee circular avatar (with profile picture or colored initials) and full name (e.g. <code>Uttam Kumar</code> , <code>Ajay Reynolds</code>).
<code>ATTENDANCE DATE</code>	Date (<code>Medium Format</code>)	Displays the target shift date requested for correction (e.g. <code>Aug 17, 2026</code>).
<code>REQUESTED CHECK IN</code>	12-Hour Time (Green text)	Displays requested check-in timestamp (e.g. <code>10:00 AM</code>) or <code>-</code> if unadjusted.
<code>REQUESTED CHECK OUT</code>	12-Hour Time (Green text)	Displays requested check-out timestamp (e.g. <code>07:30 PM</code>) or <code>-</code> if unadjusted.
<code>REASON</code>	Text string	Displays the employee's submitted explanation for the correction (e.g. <code>gdfgdgrgdr</code> , <code>sjhfvsjdui</code>).
<code>STATUS</code>	Status Chip Badge	Displays yellow <code>Pending</code> chip badge (<code>#FEF3C7</code> background) for requests awaiting review.
<code>ACTIONS</code>	Action Icon Button	Renders a blue square review icon button (<code><RateReviewOutlinedIcon /></code>). Hovering displays tooltip <code>"Review Request"</code> . Clicking opens the Review Regularization Request dialog.

2. HR Review Regularization Request Modal (`ReviewRegularizationDialog.tsx`)

ELEMENT / FIELD LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	HR REVIEWER BEHAVIOR & EXECUTION LOGIC
Employee & Attendance Details	Info Card Grid	Display Only	Displays a shaded summary card containing: • Employee: Full name and employee code in parentheses (e.g. <code>Uttam Kumar (EMP-007)</code>). • Attendance Date: Shift date being regularized (e.g. <code>Aug 17, 2026</code>). • Requested Check In: Green text timestamp (e.g. <code>10:00 AM</code>). • Requested Check Out: Green text timestamp (e.g. <code>07:30 PM</code>). • Reason for Correction: Italicized quote of employee's reason (<code>"gdfgdgrgdr"</code>).
Reviewer Comments	Multiline Text Field (<code>rows={3}</code>)	Optional	Text area with placeholder <code>"Provide comments for approval or rejection..."</code> . Allows HR to type notes for approval verification or rejection justification.
<code>[Cancel]</code> Button	Secondary Text Button	Click Action	Dismisses the review dialog without saving or making changes to the request.

ELEMENT / FIELD LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	HR REVIEWER BEHAVIOR & EXECUTION LOGIC
[Reject Request] Button	Outlined Danger Button (Red #FCA5A5)	Click Action	Sends <code>PATCH /api/v1/attendance/regularizations/:id/review</code> with <code>status: "REJECTED"</code> and reviewer comments. Updates request status chip to <code>Rejected</code> (Red) and closes dialog. Attendance record remains unchanged.
[Approve Request] Button	Solid Success Button (Green #10B981)	Click Action	Sends <code>PATCH /api/v1/attendance/regularizations/:id/review</code> with <code>status: "APPROVED"</code> and reviewer comments. Automatic Backend Effects on Approval: 1. Injects manual session punch logs (<code>SessionType.CHECK_IN</code> / <code>CHECK_OUT</code> , <code>PunchSource.MANUAL</code>). 2. Recalculates total <code>workedMinutes</code> from session pairings. 3. Recalculates attendance status (<code>PRESENT</code> , <code>LATE</code> , <code>HALF_DAY</code> , <code>isLate</code> , and <code>isCheckOutEarly</code>). 4. Marks attendance record with <code>isRegularized: true</code> . 5. Updates request status to <code>Approved</code> (Green) and refreshes queue table.

3. Employee Request Submission Dialog (`RegularizeRequestDialog.tsx` — Reference)

(Note: Used by Employees to submit requests prior to HR review)

- **Check-In / Check-Out Checkboxes:** Selects which punch time to adjust (`Requested Check-In Time` / `Requested Check-Out Time`).
- **Reason for Regularization:** Multiline text input requiring minimum 10 characters explaining the correction requirement.

✅ Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Load HR Review Queue	1. Log in as HR Admin or Manager. 2. Click Regularization in left sidebar.	Navigates to <code>/attendance/regularizations</code> . Header displays <code>"Attendance Regularizations"</code> . Table lists all pending regularization requests with employee names, dates, requested times, reasons, and yellow <code>Pending</code> chips.	<code>RegularizationListPage.tsx:L73-L99</code>
Open Review Dialog	1. Locate employee row (e.g. <code>Uttam Kumar</code>). 2. Click the blue Review Request icon button under ACTIONS.	Opens Review Regularization Request modal. Summary card displays employee code <code>Uttam Kumar (EMP-007)</code> , requested times <code>10:00 AM / 07:30 PM</code> , and reason <code>"gdfgdgrgdr"</code> .	<code>ReviewRegularizationDialog.tsx:L95-L140</code>
Approve Request (HR Action)	1. Inside Review modal, enter optional Reviewer Comments (e.g. <code>"Approved after shift verification"</code>). 2. Click <code>[Approve Request]</code> (Solid Green button).	Modal closes. Request status updates to <code>Approved</code> (Green chip). Backend automatically injects manual punch sessions, recalculates worked hours, updates shift status, and sets <code>isRegularized: true</code> .	<code>ReviewRegularizationDialog.tsx:L40-L54</code> , <code>regularization.service.ts:L95-L156</code>

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Reject Request (HR Action)	1. Inside Review modal, enter Reviewer Comments (e.g. "Biometric log mismatch"). 2. Click [Reject Request] (Outlined Red button).	Modal closes. Request status updates to Rejected (Red chip). Original employee attendance record remains untouched.	<code>ReviewRegularizationDialog.tsx:L40-L54</code>
Manual Queue Refresh	1. Click top-right [Refresh] button.	Queue re-fetches immediately, updating table entries and clearance status.	<code>RegularizationListPage.tsx:L174-L191</code>
Auto Queue Polling	1. Leave page open on HR dashboard.	Queue automatically polls backend every 15 seconds and refetches whenever browser window regains focus, introducing new employee requests in real time.	<code>RegularizationListPage.tsx:L115-L127</code>

⚠ Validation Errors to Test

FIELD / CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Already Reviewed Request (400)	Submitting an approval or rejection for a request that was already reviewed in another session	Red inline Alert displays: <code>"This request has already been reviewed"</code>	<code>regularization.service.ts:L84-L86</code>
No Time Selected (Employee Submission)	Employee submits regularization without checking Check-In or Check-Out checkbox	Red inline Alert displays: <code>"Please select at least one check-in or check-out time to adjust."</code>	<code>RegularizeRequestDialog.tsx:L87-L90</code>
Short Reason (Employee Submission)	Employee enters reason with fewer than 10 characters	Red inline Alert displays: <code>"Please provide a detailed reason (minimum 10 characters)."</code>	<code>RegularizeRequestDialog.tsx:L92-L95</code>
Duplicate Pending Request (409)	Employee submits a second request for an attendance record already pending review	Red inline Alert displays: <code>"You already have a pending request for this attendance record"</code>	<code>regularization.service.ts:L47-L50</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Request Not Found (404)	Regularization request was deleted before HR review	Red inline Alert displays: <code>"Regularization request not found"</code> .	<code>regularization.service.ts:L83</code>

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Server Review Error (500)	Database failure during review submission	Red inline Alert displays: <code>err.response.data.message</code> or <code>"Failed to submit review"</code> .	<code>ReviewRegularizationDialog.tsx:L55-L60</code>
Empty Pending Queue	No regularization requests require HR review	Displays empty queue state card with icon <code><HistoryEduOutlinedIcon /></code> , title <code>"All Caught Up!"</code> , and subtext <code>"There are no pending regularization requests requiring your review."</code>	<code>RegularizationListPage.tsx:L220-L228</code>

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS QUEUE?	HR REVIEW ACTIONS & ACCESS RIGHTS	CODE REFERENCE
<code>HR_ADMIN</code>	✓ Full HR Access	Full access to pending regularization queue (<code>getPendingRequests</code>). Can review, approve, or reject any employee request.	<code>RegularizationListPage.tsx:L73-L97</code>
<code>ORG_ADMIN</code>	✓ Full Access	Full access to pending regularization queue (<code>getPendingRequests</code>). Can review, approve, or reject requests across all branches.	<code>RegularizationListPage.tsx:L73-L97</code>
<code>BRANCH_ADMIN</code>	✓ Branch Scope	Views pending regularization queue filtered to their assigned branch (<code>findPendingForBranch</code>). Can review, approve, or reject branch requests.	<code>RegularizationListPage.tsx:L73-L97</code> , <code>regularization.service.ts:L77</code>
<code>LEADERSHIP</code>	✓ Review Access	Views pending regularization queue and can review employee requests.	<code>RegularizationListPage.tsx:L73-L97</code>
<code>MANAGER</code>	✓ Team Scope	Views pending regularization queue for team members (<code>findPendingForBranch</code>). Can review, approve, or reject requests.	<code>RegularizationListPage.tsx:L73-L97</code>
<code>PRODUCT_MANAGER</code>	✓ Review Access	Views pending regularization queue and can review requests.	<code>RegularizationListPage.tsx:L73-L97</code>
<code>EMPLOYEE</code>	✗ Review Disabled	Views only self-submitted request history (<code>getMyRegularizationRequests</code>). Review icon buttons are hidden and replaced with REQUESTED ON dates. Cannot approve or reject requests.	<code>RegularizationListPage.tsx:L65-L72</code> , <code>RegularizationListPage.tsx:L495-L499</code>

5. Leave & Approvals Management

- Routes: `/leave` , `/leave/approvals`

5.1 Leave Dashboard & Request Submission

- **Route:** `/leave`
- **Source Component:** `LeaveDashboardView.tsx`, `ApplyLeaveDialog.tsx`, `LeaveBalancesGrid.tsx`, `LeaveRequestsTable.tsx`
- **API Calls:** `GET /api/v1/leave/balances/me`, `GET /api/v1/leave/types`, `POST /api/v1/leave/requests`, `GET /api/v1/leave/requests/me`, `GET /api/v1/leave/report`, `GET /api/v1/leave/comp-off/balances/me`

 **How to Get Here:** After logging in, click **Leave Management** in the left sidebar (under top navigation group, icon: `<PolicyIcon />`). This takes you to `/leave`.

- **Sidebar Menu Permission:** Guarded by `permission: "leave.read"`.
- **Employee View:** Logged in as an `EMPLOYEE` (or user without `leave.approve` / `leave.read`), renders personal leave dashboard (`<LeaveTab isViewingOther={false} user={user} />`) titled `"My Leaves"`.
- **Manager / HR / Admin View:** Logged in as HR, Manager, or Admin, renders full `<LeaveDashboardView />` with header `"Leave Management"`, `[ + Apply Leave ]` button, 4 KPI cards, and 4 navigation tabs (`Requests`, `Balances`, `Calendar`, `Policy`). If profile is incomplete (`PROFILE_INCOMPLETE_HARD`), renders Profile Verification lock screen.

 Interactive Controls, Tabs & Dialog Field Guide

The Leave Dashboard provides workforce leave analytics, leave balance grids, interactive leave calendars, and a leave application modal. Below is the field-by-field breakdown of every control, navigation tab, and modal input:

1. Navigation Tabs Bar (`LeaveDashboardView.tsx`)

TAB NAME	TAB INDEX	CONTENT & INTERACTIVE UI BEHAVIOR
<code>Requests</code>	<code>0</code> (Default)	Renders <code><LeaveRequestsTable /></code> listing all team and personal leave requests with status badges (<code>PENDING</code> , <code>APPROVED</code> , <code>REJECTED</code>), employee details, date periods, total days, and quick action buttons (<code>Approve</code> , <code>Reject</code> , <code>View Details</code>).
<code>Balances</code>	<code>1</code>	Renders <code><LeaveBalancesGrid /></code> displaying organization employee leave balance cards with stat columns for Annual , Sick , and Casual leave balances.
<code>Calendar</code>	<code>2</code>	Renders <code><LeaveCalendarView /></code> displaying an interactive monthly workforce leave calendar.
<code>Policy</code>	<code>3</code>	Renders <code><LeavePolicyView /></code> displaying organization leave policies, accrual rules, and sandwich policy details.

2. Apply Leave Dialog Modal (`ApplyLeaveDialog.tsx`)

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Leave Type	Select Dropdown (<code>TextInput select</code>)	Required	Populated dynamically from employee leave balances (<code>balances</code> prop). Displays <code>\${leaveType.name} (\${leaveType.code}) - Balance: \${available}</code> (e.g. <code>Casual Leave (CL) - Balance: 3</code>). Defaults to first available balance.
From Date	Date Input (<code>type="date"</code>), e.g. <code>2026-09-10</code>	Required	Standard ISO date picker (<code>YYYY-MM-DD</code>).
From Session	Select Dropdown (<code>TextInput select</code>)	Optional (Default: <code>Full Day</code>)	Options: <code>Full Day</code> (<code>FULL_DAY</code>), <code>First Half</code> (<code>FIRST_HALF</code>), <code>Second Half</code> (<code>SECOND_HALF</code>).

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
To Date	Date Input (<code>type="date"</code>), e.g. <code>2026-09-12</code>	Required	Standard ISO date picker (<code>YYYY-MM-DD</code>). Must be greater than or equal to <code>From Date</code> .
To Session	Select Dropdown (<code>TextInput select</code>)	Optional (Default: <code>Full Day</code>)	Options: <code>Full Day</code> (<code>FULL_DAY</code>), <code>First Half</code> (<code>FIRST_HALF</code>), <code>Second Half</code> (<code>SECOND_HALF</code>).
Reason for Leave	Multiline Text Field (<code>rows={2}</code>)	Required	Text area with placeholder <code>"Please enter a detailed reason (minimum 5 characters) ..."</code> . Minimum 5 characters required. Displays live error helper string if 1 to 4 chars typed (<code>"Reason must be at least 5 characters long."</code>).
<code>[Submit]</code> Button	Primary Action Button (Indigo <code>#6D5DF6</code>)	Click Action	Disabled when: <code>submitting</code> , <code>!leaveTypeId</code> , <code>!fromDate</code> , <code>!toDate</code> , or <code>reason.trim().length < 5</code> . Behavior: Dispatches <code>applyLeaveRequest()</code> , sets spinner (<code>submitting: true</code>), closes modal on success, and reloads leave balances.
<code>[Cancel]</code> Button	Secondary Button (Text button)	Click Action	Dismisses the dialog without submitting changes.

3. Leave Request Details Modal (`LeaveDetailDialog`)

- **Trigger:** Clicking **View Details** or preview row on `<LeaveRequestsTable />` .
- **Content:** Displays Employee Name, Leave Type Name, Date Period & Total Days, Reason, and Status badge.

✅ Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Open Apply Leave Dialog	1. Open <code>/leave</code> 2. Click <code>[+ Apply Leave]</code> button in page header.	Dialog opens titled <code>"Apply for Leave"</code> . Leave Type dropdown populates with active balances (e.g. <code>Casual Leave (CL) - Balance: 3</code>).	<code>LeaveDashboardView.tsx:L243-L246</code> , <code>ApplyLeaveDialog.tsx:L77-L126</code>
Submit Leave Request	1. Select Leave Type: <code>Casual Leave</code> . 2. Select From Date: <code>2026-09-10</code> , To Date: <code>2026-09-12</code> . 3. Enter Reason: <code>"Family function in native place"</code> . 4. Click <code>[Submit]</code> .	Modal closes. Redux dispatches <code>applyLeaveRequest</code> . Success toast appears, leave balances reload, and request enters <code>Leave Requests</code> table with status <code>PENDING</code> .	<code>ApplyLeaveDialog.tsx:L56-L67</code> , <code>LeaveDashboardView.tsx:L228-L241</code>

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Switch Navigation Tabs	1. Click Balances tab. 2. Click Calendar tab. 3. Click Policy tab.	Tab indicator glides to selected tab. Renders <code><LeaveBalancesGrid /></code> with employee quota stat columns, <code><LeaveCalendarView /></code> monthly calendar grid, and <code><LeavePolicyView /></code> rule cards respectively.	<code>LeaveDashboardView.tsx:L476-L536</code>
Quick Approve Leave Request	1. On <code>Requests</code> tab, locate pending request. 2. Click Approve action button.	Snackbar alert displays <code>"Leave request approved for [Employee Name]"</code> . Request status updates to <code>APPROVED</code> (Green) and backend confirms balance deduction.	<code>LeaveDashboardView.tsx:L275-L297</code>
Preview Request Details	1. Click row or details icon on any leave request.	Opens <code>LeaveDetailDialog</code> modal showing employee name, leave type, period, total days, reason text, and colored status badge.	<code>LeaveDashboardView.tsx:L59-L111</code>

⚠ Validation Errors to Test

FIELD / CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Short Reason (< 5 Chars)	Enter 1 to 4 characters in Reason field	Helper error string appears below input: <code>"Reason must be at least 5 characters long."</code> . Submit button remains disabled.	<code>ApplyLeaveDialog.tsx:L189</code>
Date Order Constraint (<code>CreateLeaveRequestDto</code>)	Select <code>To Date</code> prior to <code>From Date</code> (e.g. From: <code>2026-09-10</code> , To: <code>2026-09-05</code>)	Backend Zod/Service validation fails: <code>"toDate cannot be before fromDate"</code>	<code>leave-request.service.ts:L78</code>
Advance Notice Violation	Apply for a leave type requiring notice without meeting notice days	Backend validation fails: <code>"This leave type requires at least [X] days advance notice"</code>	<code>leave-request.service.ts:L84-L87</code>
Consecutive Days Limit	Request days exceeding <code>maxConsecutiveDays</code> limit for selected leave type	Backend validation fails: <code>"Maximum [X] consecutive days allowed for this leave type"</code>	<code>leave-request.service.ts:L103-L106</code>

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Hard Profile Lock (403)	User profile is incomplete (<code>PROFILE_INCOMPLETE_HARD</code>)	<code>useProfileBlockDetect</code> catches block error (<code>isProfileBlocked: true</code>). Page replaces leave dashboard with Profile Verification card featuring <code><LockOutlinedIcon /></code> , title "Access Restricted – Profile Verification Required", pending section chips, and a <code>[Go to Profile Setup]</code> button linking to <code>/onboarding</code> .	<code>LeaveDashboardView.tsx:L394-L464</code>
Unlinked Employee Account (404)	User account has no linked <code>employeeId</code> record in database	Backend returns 404 error: "No employee record is linked to this account". Red inline Alert banner displays error in dialog.	<code>leave-request.service.ts:L59</code>
Server Error on Submit (500)	Backend DB connection error during leave application	Red inline Alert displays in dialog: <code>error</code> message string.	<code>ApplyLeaveDialog.tsx:L104-L108</code>

🔑 Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS /LEAVE ?	ACCESS SCOPE & PAGE BEHAVIOR	CODE REFERENCE
<code>ORG_ADMIN</code>	✅ Full Access	Full access to <code>/leave</code> . Views organization-wide leave requests (<code>getLeaveReport</code>), employee balances grid, and policy cards. Note: <code>[+ Apply Leave]</code> button is hidden for <code>ORG_ADMIN</code> (<code>role !== "ORG_ADMIN"</code>).	<code>LeaveDashboardView.tsx:L370</code>
<code>HR_ADMIN</code>	✅ Full Access	Full access to <code>/leave</code> . Can click <code>[+ Apply Leave]</code> , view team requests, manage balances, and approve/reject leave requests (<code>canApproveLeaves: true</code>).	<code>LeaveDashboardView.tsx:L370-L390</code>
<code>BRANCH_ADMIN</code>	✅ Branch Scope	Access to <code>/leave</code> . Can click <code>[+ Apply Leave]</code> , view branch employee leave requests, and approve/reject branch requests.	<code>LeaveDashboardView.tsx:L370-L390</code>
<code>LEADERSHIP</code>	✅ Read & Approve	Access to <code>/leave</code> . Can view leave requests, balances, calendar, and approve/reject team leave applications.	<code>LeaveDashboardView.tsx:L134-L136</code>
<code>MANAGER</code>	✅ Team Scope	Access to <code>/leave</code> . Can apply for self leave, view team leave requests, and approve/reject team leave applications (<code>canApproveLeaves: true</code>).	<code>LeaveDashboardView.tsx:L134-L136</code>
<code>PRODUCT_MANAGER</code>	✅ Permission-Based	Access determined by <code>leave.read</code> and <code>leave.approve</code> permissions.	<code>LeaveDashboardView.tsx:L134-L136</code>

ROLE SLUG	CAN ACCESS <code>/LEAVE</code> ?	ACCESS SCOPE & PAGE BEHAVIOR	CODE REFERENCE
<code>EMPLOYEE</code>	☒ Self Service	Page renders personal leave view (<code><LeaveTab isViewingOther={false} user={user} /></code>) titled <code>"My Leaves"</code> . Can view personal leave balances and submit self leave applications. Cannot view organization-wide leave reports or approve team requests.	<code>LeaveDashboardView.tsx:L349-L359</code>

5.2 Leave Approvals Queue (Manager View)

- **Route:** `/leave/approvals`
- **Source Component:** `LeaveApprovalsView.tsx` , `leave-request.service.ts`
- **API Calls:** `GET /api/v1/leave/requests/pending` , `PATCH /api/v1/leave/requests/:id/review`

 **How to Get Here:** After logging in as a **Manager**, **HR Admin**, **Branch Admin**, or **Org Admin**:

1. Click **Leave Management** in the left sidebar (under top navigation group, icon: `<PolicyIcon />`).
 2. On the Leave Management page, ensure the `Requests` tab is selected (default tab). This displays the team leave request table with `Approve` and `Reject` action buttons.
- **Role Access Restriction:** Users logged in as an `EMPLOYEE` (or without `leave.approve` permission) will not see team approval controls and will only see their own personal leave history.

Interactive Controls & Review Modal Field Guide

The Leave Approvals queue allows managers and administrators to review pending workforce leave applications, inspect leave durations and employee reasons, and approve or reject requests with optional reviewer notes. Below is the field-by-field breakdown of the queue table and the review dialog:

1. Pending Leave Approvals Table (`LeaveApprovalsView.tsx`)

ELEMENT / COLUMN HEADER	FORMAT & TYPE	MANAGER ACTION & INTERACTIVE UI BEHAVIOR
Page Header	Header Bar	Displays policy icon <code><PolicyOutlinedIcon /></code> , title "Leave Approvals" , and subtitle <code>"Review and process employee leave applications"</code> .
<code>Employee</code>	Text Block	Displays employee full name (e.g. <code>Uttam Kumar</code>) and ID badge code (<code>ID: EMP-007</code>).
<code>Leave Type</code>	Chip + Text	Displays a light-blue chip badge with leave type code (e.g. <code>CL</code> , <code>SL</code> , <code>EL</code>) and full name (<code>Casual Leave</code> , <code>Sick Leave</code>).
<code>Duration</code>	Date Period	Displays start date and end date formatted as <code>MMM DD, YYYY</code> (e.g. <code>Sep 10, 2026 to Sep 12, 2026</code>). Single-day leaves show only the start date.
<code>Total Days</code>	Count Badge	Displays requested day count (e.g. <code>1 Day</code> or <code>3 Days</code>).
<code>Reason</code>	Truncated Text	Displays employee's submitted explanation (e.g. <code>"Family function in native place"</code>). Hovering displays full text tooltip.

ELEMENT / COLUMN HEADER	FORMAT & TYPE	MANAGER ACTION & INTERACTIVE UI BEHAVIOR
Approve Button	Outlined Green Button (#10B981 , icon <CheckIcon />)	Clicking opens the Approve Leave Request confirmation dialog.
Reject Button	Outlined Red Button (#EF4444 , icon <CloseIcon />)	Clicking opens the Reject Leave Request confirmation dialog.
Table Pagination	Pagination Controls	Bottom right controls (Rows per page: 5, 10, 20). Allows paging through large pending approval queues.

2. Review Request Confirmation Modal (Dialog)

ELEMENT / FIELD LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	MANAGER REVIEWER BEHAVIOR & BACKEND MECHANICS
Modal Title	Header Title	Display Only	Displays "Approve Leave Request" (Green) or "Reject Leave Request" (Red) based on the clicked action.
Confirmation Prompt	Body Text	Display Only	Displays prompt: "Are you sure you want to approve/reject this leave request? You can add review comments below."
Review Comments	Multiline Text Field (rows={3})	Optional	Text area with placeholder "Add comments or notes...". Accepts up to 500 characters of manager review feedback.
[Cancel] Button	Text Button	Click Action	Dismisses the dialog without submitting changes or altering the request status.
[Confirm Approve] / [Confirm Reject] Button	Solid Action Button (Green #10B981 / Red #EF4444)	Click Action	Submits PATCH /api/v1/leave/requests/:id/review with payload { status: "APPROVED"

✓ Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Load Pending Approvals Queue	1. Log in as Manager or HR Admin . 2. Navigate to /leave/approvals .	Page header displays "Leave Approvals" . Table populates with all pending employee leave requests displaying names, codes, leave types, date ranges, total days, and reasons.	LeaveApprovalsView.tsx:L136-L193
Open Review Dialog	1. Locate employee row (e.g. Uttam Kumar). 2. Click Approve or Reject button.	Opens confirmation modal. Title displays "Approve Leave Request" or "Reject Leave Request" . Includes prompt and Review Comments textarea.	LeaveApprovalsView.tsx:L117-L120 , LeaveApprovalsView.tsx:L322-L372
Approve Leave Request	1. Open Review modal for approval. 2. Enter optional comment (e.g. "Approved by Manager"). 3. Click [Confirm Approve] .	Modal closes. Request disappears from pending queue. Backend converts reserved balance to used , marks daily attendance as ON_LEAVE , and logs audit action.	LeaveApprovalsView.tsx:L121-L132 , leave-request.service.ts:L283-L293

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Reject Leave Request	1. Open Review modal for rejection. 2. Enter optional comment (e.g. "Project deliverable deadline conflict"). 3. Click [Confirm Reject].	Modal closes. Request status updates to <code>REJECTED</code> . Backend releases reserved leave days back to employee's available pool.	LeaveApprovalsView.tsx:L121-L132 , leave-request.service.ts:L275-L281
Paginate Pending Queue	1. Click page navigation arrows or change rows per page dropdown (e.g. <code>20</code>).	Table fetches next page of pending leave applications (<code>getPendingLeaveRequestsRequest</code>).	LeaveApprovalsView.tsx:L308-L316

⚠ Validation Errors to Test

FIELD / CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Already Processed Request (400)	Submitting review for a request already approved/rejected in another session	Red Alert inside modal: "This request is not pending review"	leave-request.service.ts:L245
Review Comments Length (> 500 Chars)	Enter review comments exceeding 500 characters	Backend DTO validation fails: "String must contain at most 500 character(s)"	leave.dto.ts:L56

✖ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Unauthorized Approver Role (403)	Manager attempts to approve a level reserved for a different role	Red inline Alert displays: "Only a [Role] can act on this approval level" .	leave-request.service.ts:L254-L257
Request Not Found (404)	Leave request was deleted before review submission	Red inline Alert displays: "Leave request not found" .	leave-request.service.ts:L243
Empty Approvals Queue	No pending employee leave requests require review	Renders empty state card titled "No Pending Leave Requests" , subtext "Excellent! There are no employee leave applications awaiting your review."	LeaveApprovalsView.tsx:L162-L178

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS <code>/LEAVE/APPROVALS</code> ?	ACCESS SCOPE & REVIEW RIGHTS	CODE REFERENCE
<code>ORG_ADMIN</code>	✓ Full Access	Full access to <code>/leave/approvals</code> (bypasses step approver role checks). Can review and approve/reject leave requests for any employee across all branches.	<code>leave-request.service.ts:L253</code>
<code>HR_ADMIN</code>	✓ Full Access	Full access to <code>/leave/approvals</code> (<code>permission: "leave.approve"</code>). Can review, approve, or reject employee leave applications.	<code>index.tsx:L150-L156</code>
<code>BRANCH_ADMIN</code>	✓ Branch Scope	Access to <code>/leave/approvals</code> . Can review, approve, or reject leave requests within assigned branch.	<code>index.tsx:L150-L156</code>
<code>LEADERSHIP</code>	✓ Review Access	Access to <code>/leave/approvals</code> . Can review and process pending team leave applications.	<code>index.tsx:L150-L156</code>
<code>MANAGER</code>	✓ Team Scope	Access to <code>/leave/approvals</code> (<code>permission: "leave.approve"</code>). Can review and process pending leave requests for direct report team members.	<code>index.tsx:L150-L156</code>
<code>PRODUCT_MANAGER</code>	✓ Permission-Based	Access granted if assigned <code>permission: "leave.approve"</code> .	<code>index.tsx:L150-L156</code>
<code>EMPLOYEE</code>	✗ Blocked	Route <code>/leave/approvals</code> is blocked by <code><RoleGuard permission="leave.approve"></code> . Employees are redirected away from the approvals queue.	<code>index.tsx:L150-L156</code>

6. 📅 Holidays & Branch Calendars

- **Route:** `/holidays`
- **Source Component:** `HolidayListView.tsx`, `holiday.service.ts`, `holiday-resolution.engine.ts`
- **API Calls:** `GET /api/v1/leave/holidays?year=2026`, `POST /api/v1/leave/holidays`, `PATCH /api/v1/leave/holidays/:id`, `DELETE /api/v1/leave/holidays/:id`, `GET /api/v1/leave/holidays/resolve?branchId=:id&year=2026`, `POST /api/v1/leave/holidays/seed-default`

🗺️ **How to Get Here:** After logging in:

1. Click **Holiday** in the left sidebar (under top navigation group, icon: `<CalendarMonthIcon />`). This opens the **Holidays & Branch Calendars** screen (`/holidays`).
2. On the Holidays page, use the top action buttons `[ Seed Defaults ]` and `[ Add Holiday ]` (visible to Admin/HR roles) or use the Filter Toolbar to switch between master lists, branch-resolved schedules, monthly calendar grids, and personal schedules.

- **Sidebar Menu Permission:** Guarded by `permission: "leave.read"`.

Interactive Controls, Filter Bar & Form Field Guide

The Holidays module provides organization-wide statutory holiday management, priority resolution engines (`BRANCH > STATE > COUNTRY > GLOBAL`), interactive monthly branch calendars, and statutory defaults seeding. Below is the field-by-field breakdown of every control, filter, and modal dialog:

1. Top Filter Toolbar Controls (`HolidayListView.tsx`)

ELEMENT / CONTROL LABEL	FORMAT & TYPE	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Year Selector	Select Dropdown (<code>TextInput select</code>)	Options: <code>2024</code> , <code>2025</code> , <code>2026</code> , <code>2027</code> , <code>2028</code> . Defaults to current year (<code>2026</code>). Selecting a year refetches holidays for that year (<code>listHolidaysRequest(selectedYear)</code>).
View Mode Selector	Select Dropdown (<code>TextInput select</code>)	Visible to authorized roles (<code>ORG_ADMIN</code> , <code>HR_ADMIN</code> , <code>isSuperAdmin</code>). Options: <code>All Organization Master Holidays</code> (<code>ALL</code> , Default): Renders master holiday table. <code>Resolve Branch List</code> (<code>RESOLVED</code>): Displays priority-deduplicated branch list. <code>Branch Monthly Calendar Grid</code> (<code>GRID</code>): Displays interactive month grid (<code>BranchCalendarGrid</code>). <code>My Personal Schedule</code> (<code>MY_SCHEDULE</code>): Displays employee's personal schedule calendar.
Branch Selector	Select Dropdown (<code>TextInput select</code>)	Visible when <code>viewModel</code> is set to <code>RESOLVED</code> or <code>GRID</code> . Populated dynamically from organization branches (<code>listBranches()</code>). Selecting a branch refetches resolved branch holidays or monthly calendar.

2. Add / Edit Holiday Dialog Modal (`HolidayFormDialog`)

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Holiday Name	Text Field (<code>TextInput</code>)	Required	Text input with placeholder "e.g. Republic Day or Bangalore Office Day Off" . E.g. <code>Republic Day</code> , <code>Diwali</code> .
Date	Date Input (<code>type="date"</code>)	Required	Standard ISO date picker (<code>YYYY-MM-DD</code>).
Holiday Type	Select Dropdown (<code>TextInput select</code>)	Optional (Default: <code>NATIONAL</code>)	Options: <code>National Holiday</code> (<code>NATIONAL</code> , Red chip <code>#FEE2E2</code>), <code>Restricted Holiday</code> (<code>RESTRICTED</code> , Yellow chip <code>#FEF3C7</code>), <code>Regional Holiday</code> (<code>REGIONAL</code> , Blue chip <code>#E0F2FE</code>).
Holiday Scope	Select Dropdown (<code>TextInput select</code>)	Optional (Default: <code>GLOBAL</code>)	Options: <code>Global (Entire Organization / Tenant)</code> (<code>GLOBAL</code> , Purple chip <code>#EDE9FE</code>) <code>Country Scope</code> (<code>COUNTRY</code> , Blue chip <code>#E0F2FE</code>) <code>State Scope</code> (<code>STATE</code> , Teal chip <code>#CCFBF1</code>) <code>Branch Scope</code> (<code>BRANCH</code> , Indigo chip <code>#E0E7FF</code>).
Target Branch	Select Dropdown (<code>TextInput select</code>)	Required when Scope = <code>BRANCH</code>	Cascading Select: Appears only when <code>Holiday Scope</code> is set to <code>BRANCH</code> . Populated dynamically from active branches.

ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Country Code	Text Field (<code>TextInput</code>)	Required when Scope = <code>COUNTRY</code> or <code>STATE</code>	Cascading Field: Appears only when Scope is <code>COUNTRY</code> or <code>STATE</code> . Placeholder <code>"e.g. IN or US"</code> . 2-letter ISO country code.
State Code / State Name	Text Field (<code>TextInput</code>)	Required when Scope = <code>STATE</code>	Cascading Field: Appears only when Scope is <code>STATE</code> . Placeholder <code>"e.g. Karnataka or KA"</code> . Accepts state code or full state name.
Is Optional Holiday	MUI Switch Toggle	Optional (Default: <code>Off</code>)	Toggle label <code>"Is Optional Holiday"</code> , subtext <code>"Employees can choose to take this leave or not"</code> . Sets <code>isOptional: true</code> .
Description	Multiline Text Field (<code>rows={2}</code>)	Optional	Text area with placeholder <code>"Brief description or context"</code> .
<code>[Create]</code> / <code>[Update]</code> Button	Primary Action Button (Indigo <code>#6D5DF6</code>)	Click Action	Disabled when: <code>submitting</code> , <code>!name.trim()</code> , <code>!date</code> , or scope dependencies missing (<code>branchId</code> for <code>BRANCH</code> , <code>countryCode</code> for <code>COUNTRY</code> , <code>stateCode</code> for <code>STATE</code>). Behavior: Dispatches <code>createHolidayRequest</code> or <code>updateHolidayRequest</code> , sets spinner, closes modal, and refreshes list.
<code>[Cancel]</code> Button	Secondary Text Button	Click Action	Dismisses the dialog without saving changes.

3. Seed Default Statutory Holidays Modal (`SeedHolidaysDialog`)

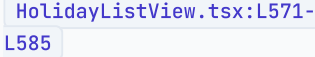
ELEMENT / CONTROL LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	SELECTION DEPENDENCIES & INTERACTIVE UI BEHAVIOR
Modal Header	Header Title	Display Only	Displays sparkle icon <code><AutoAwesomeIcon /></code> and title "Seed Default Statutory Holidays" .
Info Banner	Blue Alert Card	Display Only	Text: <code>"Generates statutory national (COUNTRY) & cantonal/regional (STATE) holidays based on your organization's registered locale. Idempotent & safe to execute."</code>
State / Canton Code	Text Field (<code>TextInput</code>)	Optional	Placeholder <code>"e.g. ZH (Zurich), GE, KA, NY, CA"</code> . Specifies regional state/canton code for statutory seeding.
<code>[Seed Statutory Holidays]</code> Button	Primary Action Button	Click Action	Submits <code>POST /api/v1/leave/holidays/seed-default?stateCode=...</code> . Automatically derives country code from organization settings.
<code>[Cancel]</code> Button	Secondary Text Button	Click Action	Dismisses the seed modal without making API calls.

4. Row Action Context Menu (`Menu`)

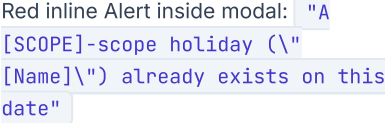
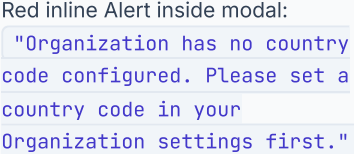
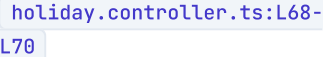
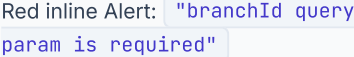
- **Trigger:** Clicking `MoreVertIcon` (`:`) on any row in the holiday table.
- **Options:**
 - `Edit` (Blue icon): Opens `HolidayFormDialog` with pre-filled details.
 - `Delete` (Red icon): Opens `ConfirmDialog` titled `"Delete Holiday"`. Clicking `Delete` dispatches `deleteHolidayRequest(id)`.

✓ Success Cases

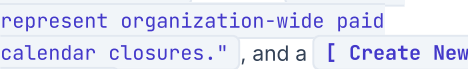
ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Load Holidays Directory	1. Log in to HRMS. 2. Click Holiday in left sidebar.	Page header displays "Holidays" . Filter toolbar and Master Holiday Table populate with active organization holidays for current year.	<code>HolidayListView.tsx:L632-L693</code>
Add New Master Holiday	1. Click <code>[Add Holiday]</code> button. 2. Enter Name: <code>Independence Day</code> , Date: <code>2026-08-15</code> , Type: <code>NATIONAL</code> , Scope: <code>GLOBAL</code> . 3. Click <code>[Create]</code> .	Modal closes. Redux dispatches <code>createHolidayRequest</code> . Toast confirms success and new holiday appears with Red <code>NATIONAL</code> chip and Purple <code>GLOBAL</code> scope chip.	<code>HolidayListView.tsx:L127-L150</code> , <code>HolidayListView.tsx:L563-L569</code>
Add Branch Local Override Holiday	1. Click <code>[Add Holiday]</code> button. 2. Select Scope: <code>BRANCH</code> . Select Branch: <code>Bangalore Branch</code> . 3. Enter Name: <code>Karnataka Rajyotsava</code> , Date: <code>2026-11-01</code> . 4. Click <code>[Create]</code> .	Target Branch select field appears. On submission, creates a branch-specific holiday with Indigo <code>BRANCH</code> scope chip. Target branch calendar automatically includes this local override.	<code>HolidayListView.tsx:L237-L251</code>
Seed Default Statutory Holidays	1. Click <code>[Seed Defaults]</code> button. 2. Optional: Enter State Code <code>KA</code> . 3. Click <code>[Seed Statutory Holidays]</code> .	Backend generates statutory national and state holidays derived from org locale (e.g. <code>IN-KA</code>). Success alert displays <code>"Statutory holidays for IN-KA seeded successfully"</code> .	<code>HolidayListView.tsx:L330-L424</code> , <code>holiday.controller.ts:L97-L123</code>
View Priority-Resolved Branch List	1. In View Mode dropdown, select <code>Resolve Branch List</code> . 2. Select Branch: <code>Bangalore Branch</code> .	Info banner displays explaining priority order (<code>BRANCH > STATE > COUNTRY > GLOBAL</code>). Table displays deduplicated holiday schedule tailored specifically for Bangalore Branch.	<code>HolidayListView.tsx:L490-L503</code> , <code>HolidayListView.tsx:L752-L757</code>
View Interactive Branch Calendar Grid	1. In View Mode dropdown, select <code>Branch Monthly Calendar Grid</code> . 2. Select Branch: <code>Bangalore Branch</code> .	Replaces table with <code><BranchCalendarGrid /></code> monthly grid showing working days, weekends, and holiday events. Use <code><</code> and <code>></code> buttons to navigate months.	<code>HolidayListView.tsx:L759-L782</code>
Edit Existing Holiday	1. Click <code>:</code> action menu on holiday row. 2. Click Edit . 3. Modify description or scope. 4. Click <code>[Update]</code> .	Modal pre-fills with existing holiday data. Submitting updates holiday and refreshes list.	<code>HolidayListView.tsx:L557-L569</code>

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Delete Holiday	1. Click  action menu on holiday row. 2. Click Delete . 3. In confirmation modal, click  .	Confirmation dialog asks  . On confirmation, deletes holiday record and updates table.	

Validation Errors to Test

FIELD / CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Duplicate Holiday Date & Scope (409 Conflict)	Creating or updating a holiday with name/date/scope matching an existing record	Red inline Alert inside modal: 	
Unconfigured Org Country Code (400)	Clicking Seed Defaults when organization locale country code is missing	Red inline Alert inside modal: 	
Invalid Year Query (400)	Providing an invalid year parameter (< 2000 or > 2100) to branch resolution API	Red inline Alert: 	
Missing Branch ID Query (400)	Accessing branch resolution without specifying branch ID	Red inline Alert: 	

Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Holiday Not Found (404)	Attempting to update or delete a holiday that was removed in another session	Red inline Alert: 	
Server Error on Create (500)	Database failure during holiday creation	Red inline Alert inside modal:  message string.	
Empty Holiday Directory	No master holidays recorded for selected year	Renders empty state paper titled  , subtext  , and a  button.	

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS <code>/HOLIDAYS</code> ?	ACCESS SCOPE & ACTIONS AVAILABLE	CODE REFERENCE
<code>ORG_ADMIN</code>	✓ Full Access	Full access to <code>/holidays</code> . Can view all master holidays, seed statutory defaults, create/edit/delete holidays, resolve branch calendars, and switch view modes.	<code>HolidayListView.tsx:L439-L440</code>
<code>HR_ADMIN</code>	✓ Full Access	Full access to <code>/holidays</code> . Can seed defaults, add/edit/delete holidays, view branch calendar grids, and resolve branch lists (<code>canCreate: true</code>).	<code>HolidayListView.tsx:L439-L440</code>
<code>BRANCH_ADMIN</code>	✓ Branch Scope	Access to <code>/holidays</code> . Can view master holidays and resolve/view calendar grid for assigned branch. <code>Add Holiday</code> and <code>Seed Defaults</code> buttons visible if granted <code>holiday.create</code> .	<code>HolidayListView.tsx:L440</code>
<code>LEADERSHIP</code>	✓ Read-Only	Access to <code>/holidays</code> . Can view master holidays and branch calendars. Action buttons (<code>Seed Defaults</code> , <code>Add Holiday</code> , Edit/Delete icons) are hidden.	<code>HolidayListView.tsx:L440</code>
<code>MANAGER</code>	✓ Read-Only	Access to <code>/holidays</code> (<code>permission: "leave.read"</code>). Can view master holidays and team branch calendars. Action buttons hidden unless granted <code>holiday.create</code> .	<code>HolidayListView.tsx:L440</code>
<code>PRODUCT_MANAGER</code>	✓ Permission-Based	Access determined by <code>leave.read</code> and <code>holiday.create</code> permissions.	<code>HolidayListView.tsx:L440</code>
<code>EMPLOYEE</code>	✓ Read-Only	Access to <code>/holidays</code> (<code>permission: "leave.read"</code>). Views master holiday table and personal schedule (<code>MY_SCHEDULE</code>). Management controls and View Mode selector are hidden (<code>isAuthorized: false</code>).	<code>HolidayListView.tsx:L720-L733</code>

7. Document Verification

- **Route:** `/document-verification`
- **Source Component:** `DocumentVerificationView.tsx`
- **API Calls:** `GET /api/v1/documents/pending`, `PATCH /api/v1/documents/:id/verify`, `GET /api/v1/documents/:empId/:docId/download`

 **How to Get Here:** After logging in as **HR Admin**, **Branch Admin**, or **Org Admin**:

1. Click **Document Verification** in the left sidebar (under bottom navigation group, icon: `<FactCheckIcon />`). This opens the **Document Verification Queue** screen (`/document-verification`).
- **Sidebar Menu Permission:** Guarded by `permission: "document.read"`.

Interactive Controls & Rejection Modal Field Guide

The Document Verification queue allows HR administrators to review employee identity and education proof documents uploaded during onboarding or profile updates:

1. Pending Verification Table (`DocumentVerificationView.tsx`)

ELEMENT / COLUMN HEADER	FORMAT & TYPE	HR ACTION & INTERACTIVE UI BEHAVIOR
Page Header	Header Bar	Displays document icon <code><DescriptionOutlinedIcon /></code> , title "Document Verification" , and subtitle "Review and verify employee documents" .
EMPLOYEE	Avatar + Text Block	Displays employee avatar initials, full name (e.g. <code>Uttam Kumar</code>), and Employee Code (<code>EMP-007</code>).
DOCUMENT TYPE	Text String	Displays uploaded document category (e.g. <code>PAN Card</code> , <code>Aadhaar Card</code> , <code>Degree Certificate</code>).
FILE NAME	Text String	Displays file name string (e.g. <code>pan_card_uttam.pdf</code>).
UPLOADED ON	Date (<code>MMM DD, YYYY</code>)	Upload timestamp.
STATUS	Status Chip Badge	Displays yellow <code>Pending</code> chip badge (<code>#FEF3C7</code> background).
View Document Button	Small Eye Icon Button (<code><VisibilityOutlinedIcon /></code>)	Clicking fetches a secure presigned download URL (<code>getHrDownloadUrl</code>) and opens the document file in a new browser tab for visual inspection.
Approve Button	Small Green Check Icon Button (<code>#DCFCE7</code> , <code><CheckIcon /></code>)	Submits <code>verifyDocument(id, { isVerified: true })</code> . On success, removes document from queue and displays toast "Document approved successfully" .
Reject Button	Small Red Close Icon Button (<code>#FEE2E2</code> , <code><CloseIcon /></code>)	Clicking opens the Reject Document dialog modal.

2. Reject Document Modal (`Dialog`)

ELEMENT / FIELD LABEL	EXPECTED FORMAT & TYPE	REQUIRED / OPTIONAL	HR REVIEWER BEHAVIOR & EXECUTION LOGIC
Modal Title	Header Title	Display Only	Displays title "Reject Document" .
Remarks Field	Multiline Text Field (<code>rows={3}</code>)	Optional	Text area label "Remarks" . Allows HR to type rejection notes or resubmission instructions for the employee.
[Reject] Button	Danger Action Button (Red <code>#DC2626</code>)	Click Action	Sends <code>PATCH /api/v1/documents/:id/verify</code> with <code>{ isVerified: false, remarks }</code> . Removes document from pending queue, updates document status to Rejected , and displays toast "Document rejected" .
[Cancel] Button	Text Button	Click Action	Dismisses the dialog without rejecting the document.

✓ Success Cases

ACTION	STEPS	EXPECTED RESULT	CODE REFERENCE
Inspect Uploaded Document	1. Open <code>/document-verification</code> . 2. Locate employee row. 3. Click View Document (Eye icon).	Fetches presigned download URL (<code>getHrDownloadUrl</code>) and opens document file in a new browser tab.	<code>DocumentVerificationView.tsx:L98-L107</code>
Approve Document	1. Locate pending document row. 2. Click Approve (Green check icon).	Document status updates to verified. Row disappears from pending queue and toast displays <code>"Document approved successfully"</code> .	<code>DocumentVerificationView.tsx:L109-L125</code>
Reject Document	1. Click Reject (Red close icon). 2. Enter Remarks: <code>"Document image blurry"</code> . 3. Click <code>[Reject]</code> .	Modal closes. Document status updates to rejected. Row disappears from queue and toast displays <code>"Document rejected"</code> .	<code>DocumentVerificationView.tsx:L133-L155</code>

⚠ Validation Errors to Test

FIELD / CONSTRAINT	TRIGGER CONDITION	EXPECTED ERROR MESSAGE	CODE REFERENCE
Missing Document ID Query	Attempting to verify or download without a valid document ID	Backend returns <code>400 Bad Request</code> or <code>404 Not Found</code> .	<code>DocumentVerificationView.tsx:L118</code>

✗ Error / Failure Cases

SCENARIO	TRIGGER CONDITION	EXPECTED BEHAVIOR	CODE REFERENCE
Server Error on Document Verification (500)	Network or database error during document approval/rejection	Red Alert banner displays: <code>"Failed to approve document"</code> or <code>"Failed to reject document"</code> .	<code>DocumentVerificationView.tsx:L118-L122</code>
Empty Verification Queue	No pending employee documents require HR review	Renders empty state card featuring <code><HourglassEmptyOutlinedIcon /></code> , title <code>"No pending documents to verify"</code> , subtext <code>"All documents have been reviewed"</code> .	<code>DocumentVerificationView.tsx:L180-L189</code>

Role-Based Access & Restrictions

ROLE SLUG	CAN ACCESS <code>/DOCUMENT-VERIFICATION</code> ?	ACCESS SCOPE & ACTIONS AVAILABLE	CODE REFERENCE
<code>ORG_ADMIN</code>	✓ Full Access	Full access to inspect, approve, or reject employee documents.	<code>index.tsx:L178-L184</code>
<code>HR_ADMIN</code>	✓ Full Access	Full access to <code>/document-verification</code> queue (<code>permission: "document.read"</code>) to review employee documents.	<code>index.tsx:L178-L184</code>
<code>BRANCH_ADMIN</code>	✓ Branch Scope	Access to <code>/document-verification</code> to review branch employee documents.	<code>index.tsx:L178-L184</code>
<code>LEADERSHIP</code>	✓ Read Access	Can view employee documents in verification queue.	<code>index.tsx:L178-L184</code>
<code>MANAGER</code>	✓ Team Scope	Access to <code>/document-verification</code> for team members if assigned <code>document.read</code> .	<code>index.tsx:L178-L184</code>
<code>PRODUCT_MANAGER</code>	✓ Permission-Based	Access determined by <code>document.read</code> permission.	<code>index.tsx:L178-L184</code>
<code>EMPLOYEE</code>	✗ Blocked	Route <code>/document-verification</code> is blocked by <code><RoleGuard permission="document.read"></code> . Employees cannot verify or reject documents.	<code>index.tsx:L178-L184</code>